Introduction to Programming

W12 Recursion

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14 January 2026
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Schedule

#	Date	Subject
1	15.10.25	Introduction
2	22.10.25	Control Flow in Programming
3	29.10.25	**In-Depth Core Concepts***
4	05.11.25	Core Data Structures
5	12.11.25	Code Reuse and Structure
6	19.11.25	Type Flexibility and Safety
7	26.11.25	**In-Depth Object Orientation***
8	03.12.25	Functional Programming Essentials
9	10.12.25	Algorithms and Data Handling
10	17.12.25	Programming Languages
11	07.01.26	Graphical User Interfaces
12	**14.01.26**	**Recursion**
13	21.01.26	Concurrency
14	28.01.26	Beyond Programming
15	04.02.26	Course Review

*** Repetition**

Roadmap

- **Context**
  - Apply OOP concepts: abstraction, encapsulation, inheritance and polymorphism
  - Use control structures and data types
  - Use enums, annotations, generics, collections, iterators, lambda expressions, and streams
  - Apply error handling, implement algorithms, and understand programming languages
  - Implement GUIs using JavaFX
- **Learning goals**
  - Apply direct and indirect recursion
  - Understand the terminating conditions of recursion
  - Implement algorithms based on **recursion** (and iteration)
  - Explain the difference between recursion and iteration
  - Transform algorithms: iteration <—> recursion

Outline

- The principle of recursion
- Example: Towers of Hanoi
- Example: Binary search
- Example: Koch snowflake
- Example: Mergesort

Divide and conquer

- Basic, yet very useful principle of algorithms
- To solve a problem
  1. Divide it into several subproblems
  2. Solve the individual subproblems
  3. Combine the partial solutions to a total solution

Recursion

- Principle of recursion: run the same calculation pattern over and over again with simpler or smaller input data until you arrive at a trivial input
- **Realization:** methods that call themselves directly or indirectly
- **Recursion** is the standard implementation of the divide and conquer principle

Recursive methods

- **Definition**: a method `f` is called (directly) recursive if calls to `f` occur in the body of `f`
- The method `f` is called (indirectly) recursive if in the body of `f` a method `g` is called, which in turn leads directly or indirectly to calls of `f`
- For each recursive call, a new instance of the respective method is started

→ Each instance has its own local variables and parameters, which are not visible from outside

Example: factorial function

- The factorial function **`n!`** calculates the product of the numbers 1, 2, …, n
- Recursively, **`n!`** can therefore be calculated as follows

$$0! = 1, \quad n! = n \cdot (n-1)! \quad (\text{for } n \geq 1)$$

Java code

```
public static long fac(long n) {
    if (n > 0) {
        return n * fac (n - 1);
    } else {
        return 1;
    }
}
```

Direct recursive method call

Important: a terminating condition without recursive function call to prevent endless program

What happens when calling **`fac`**

```
fac(4)
= ( if(4>0){ 4*fac(4-1) } else { 1 } )
= 4*fac(3) =4*( if(3>0){ 3*fac(3-1) } else{ 1 } )
= 4*3*fac(2)
= 4*3*( if (2>0) { 2*fac(2-1) } else { 1 } )
= 4*3*2*fac(1)
= 4*3*2*( if (1>0) { 1*fac(1-1) } else { 1 } )
= 4*3*2*1*fac(0)
= 4*3*2*1*( if (0>0) { 0*fac(0-1) } else { 1 } )
= 4*3*2*1*1
```

Example: wrong terminating condition

```
public static long fac(long n) {
    if (n == 100) {
        return 1;
    } else {
        return n * fac (n - 1);
    }
}
```

When you invoke this method with **n < 100**, it will not terminate

```
Exception in thread "main" java.lang.StackOverflowError
```

Example: greatest common divisor

```
int gcd(int n, int m) {
    if (n == m) {
        return n;
    } else if (n < m) {
        return gcd(n, m - n);
    } else {
        return gcd(n - m, m);
    }
}
```

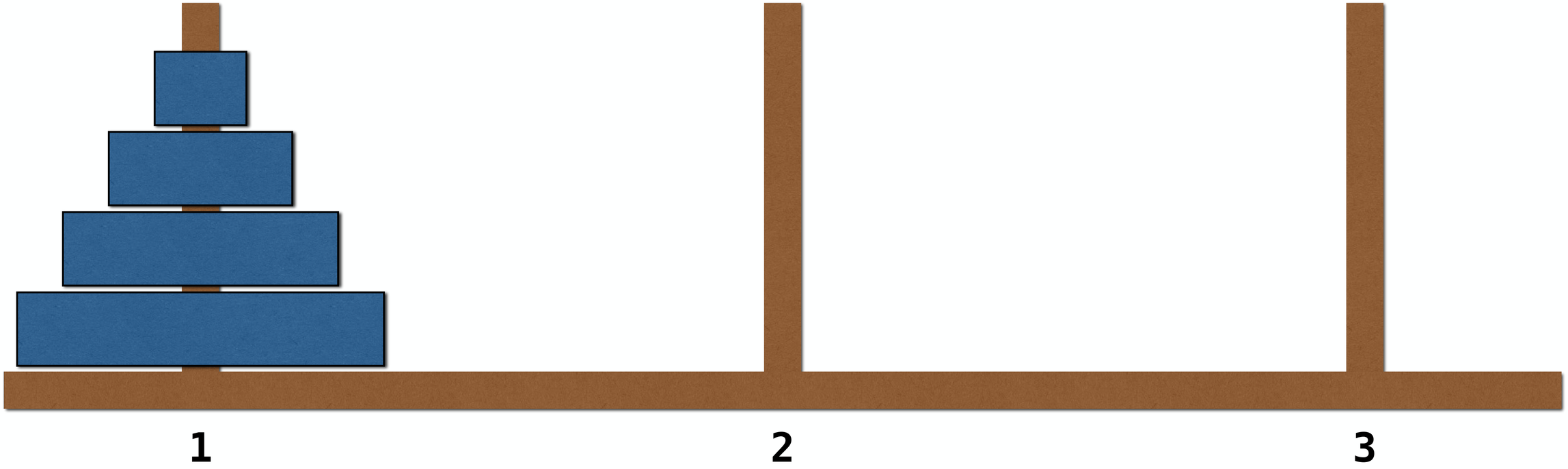
Important: a terminating condition without recursion to prevent endless program

Direct recursive method calls

Outline

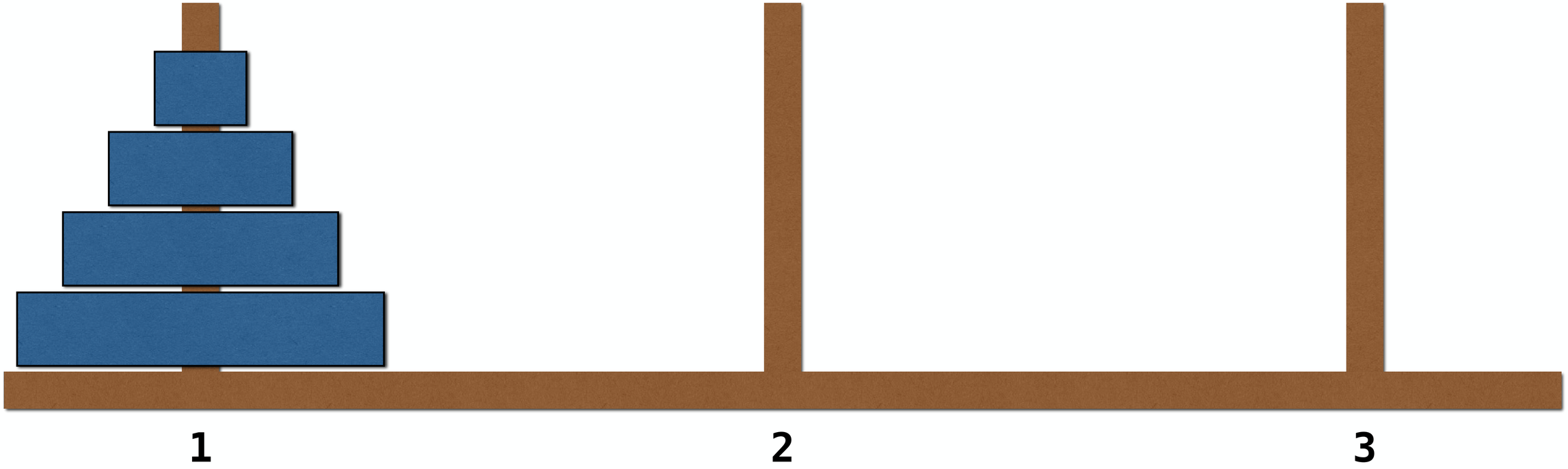
- The principle of recursion
- Example: Towers of Hanoi
- Example: Binary search
- Example: Koch snowflake
- Example: Mergesort

Problem



- Move the stack from left to right
- You may move exactly one ring in each turn
- A larger ring may never be placed on top of a smaller one

Move slices



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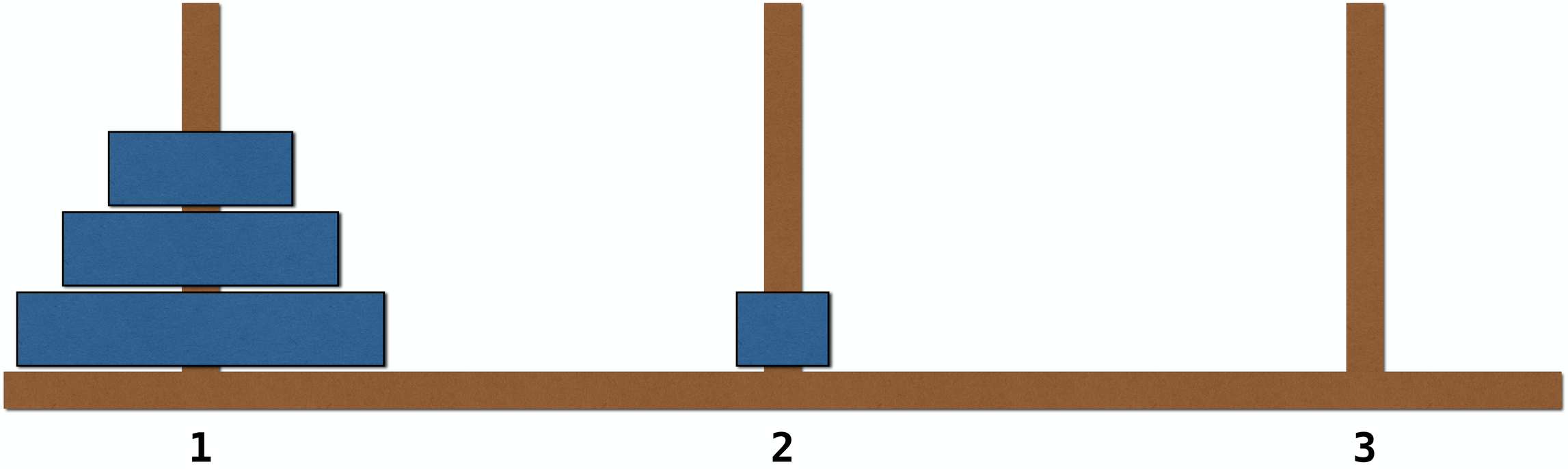

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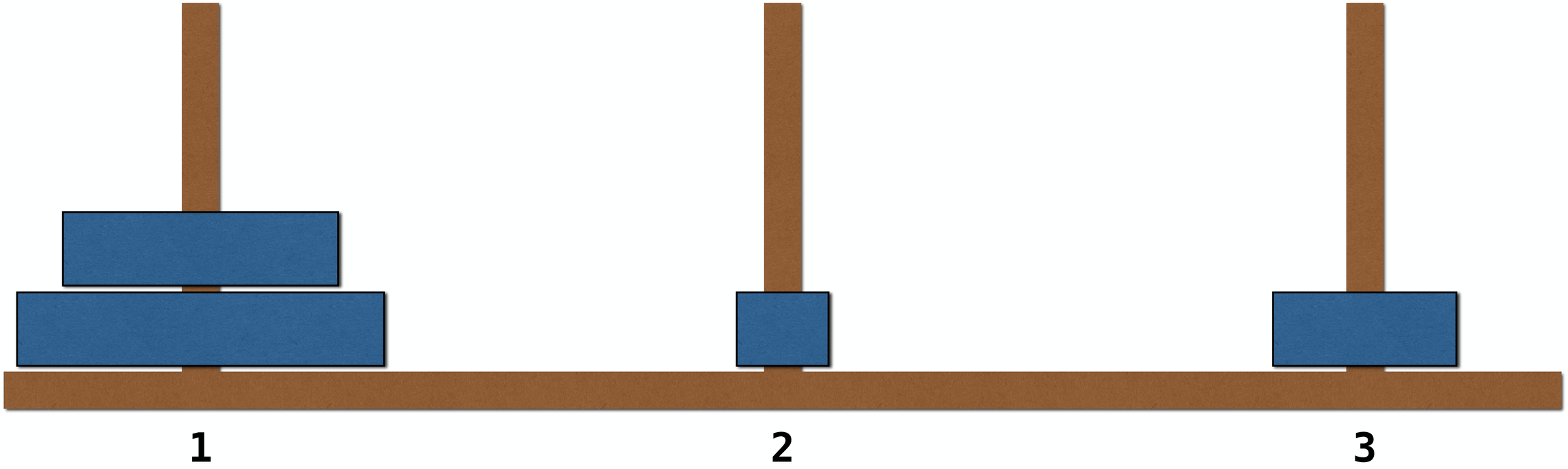

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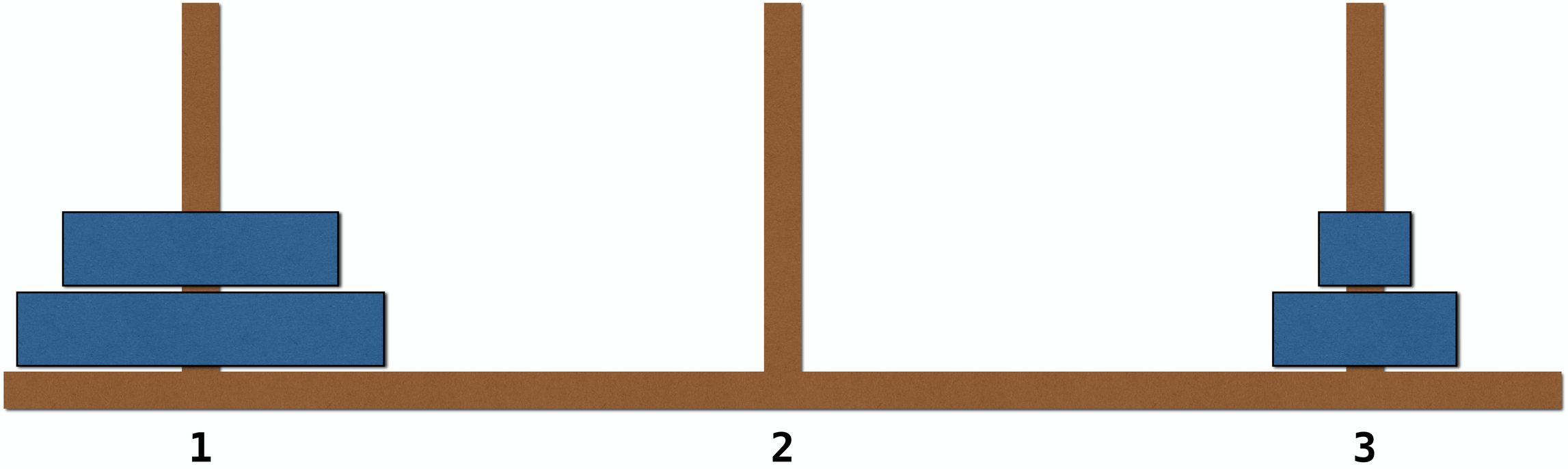

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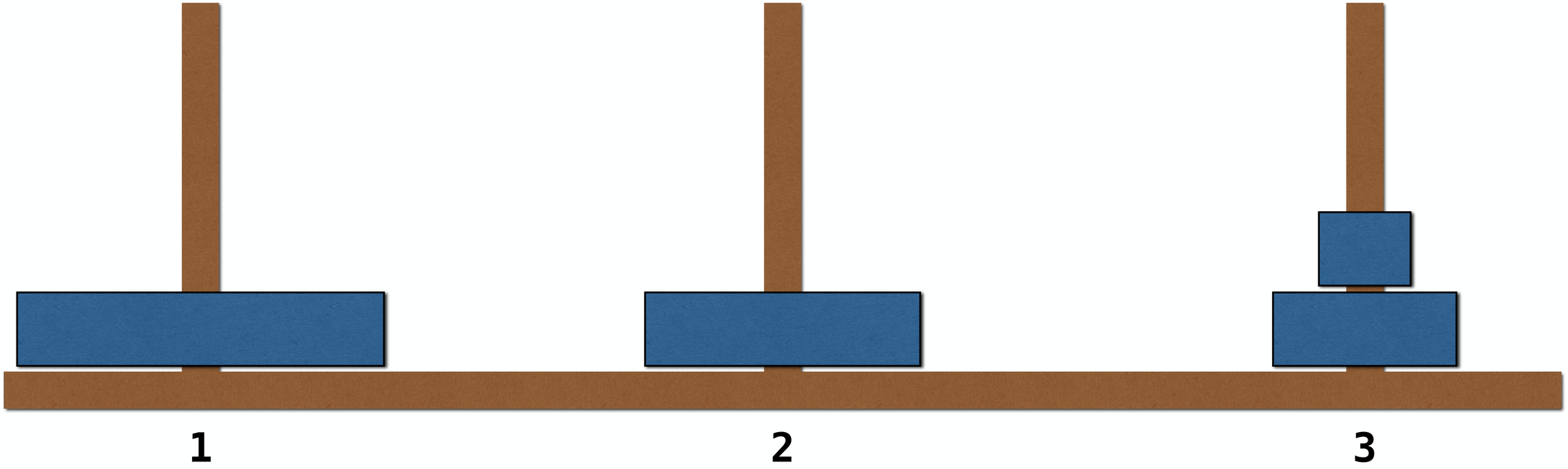

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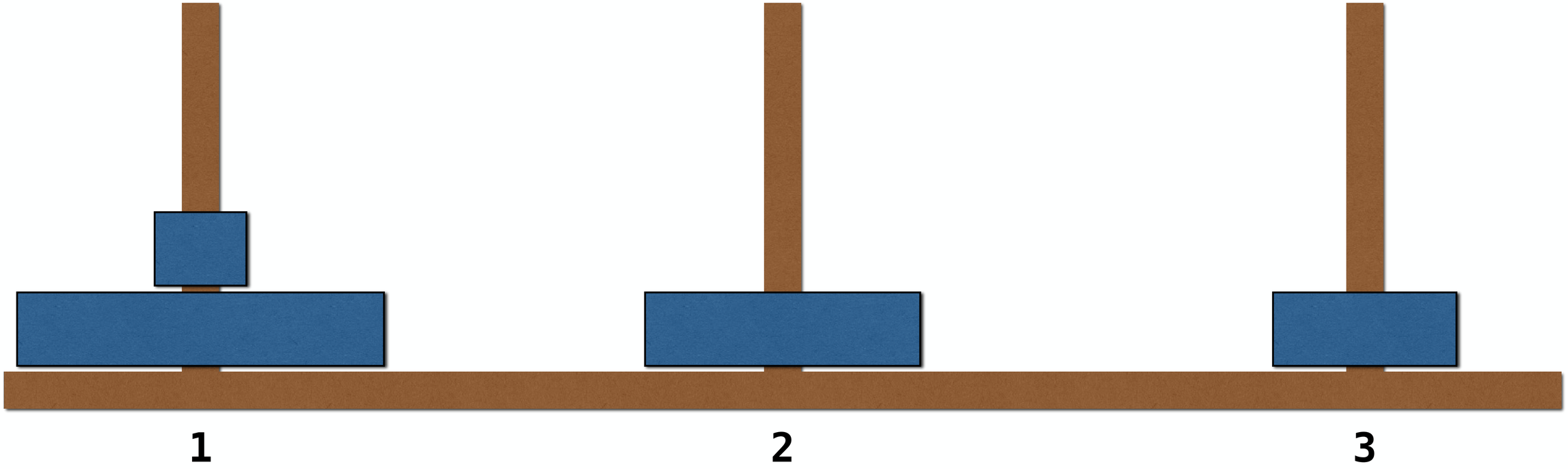

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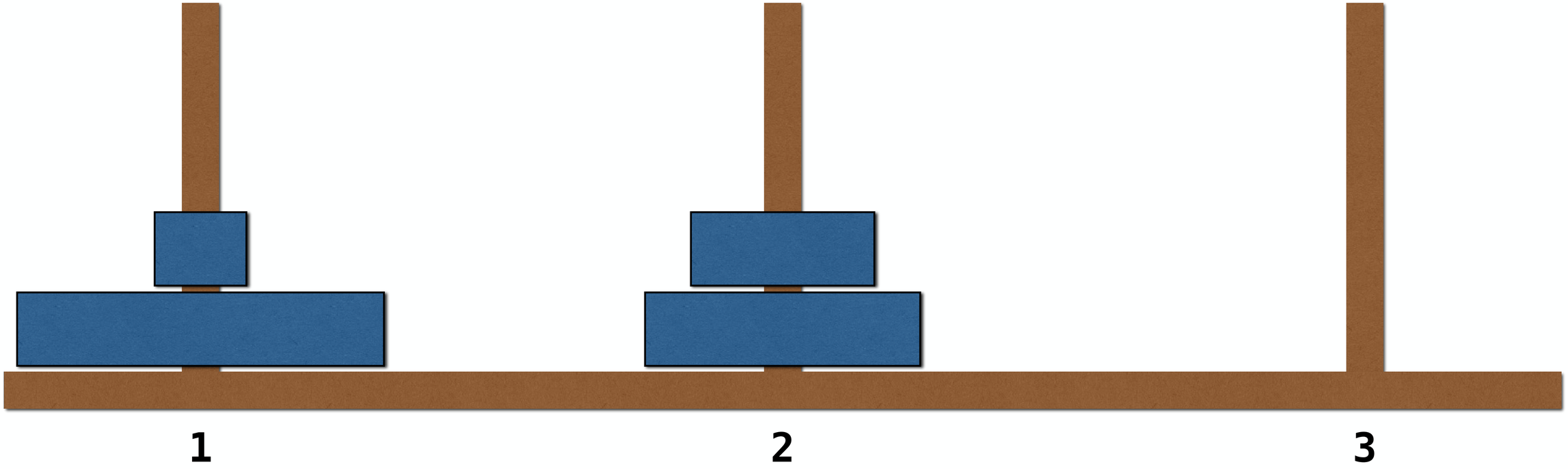

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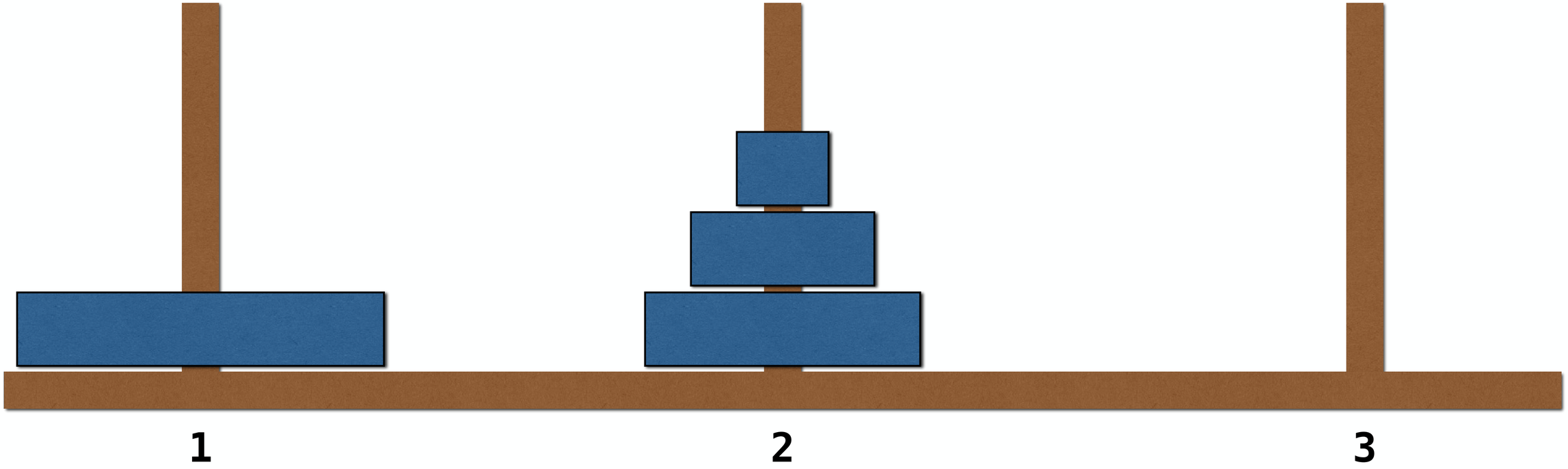

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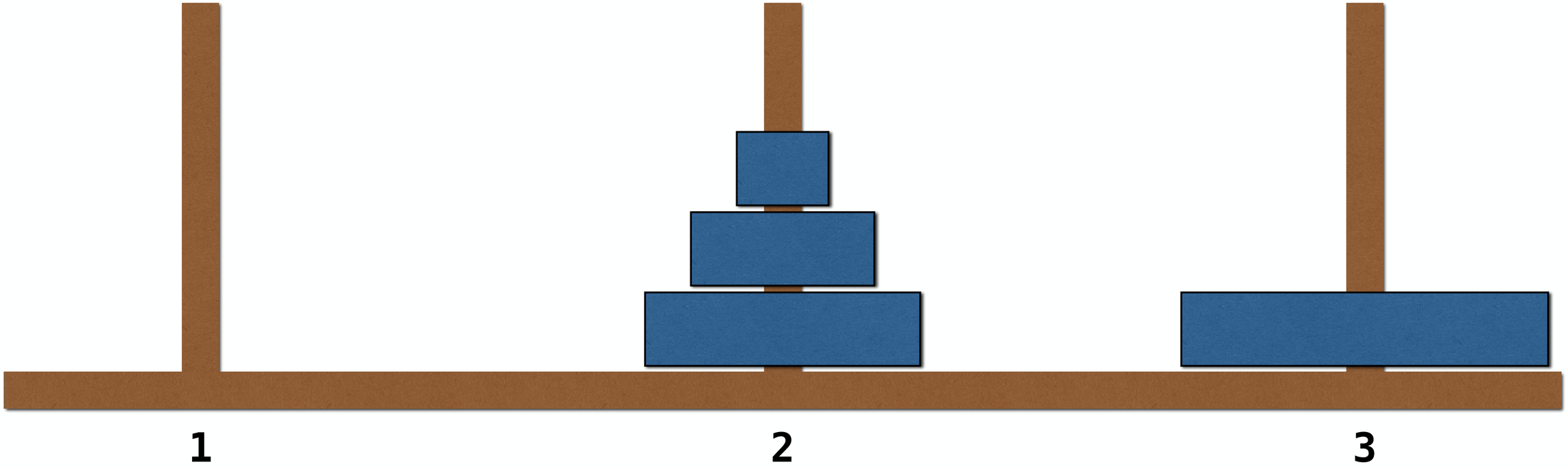

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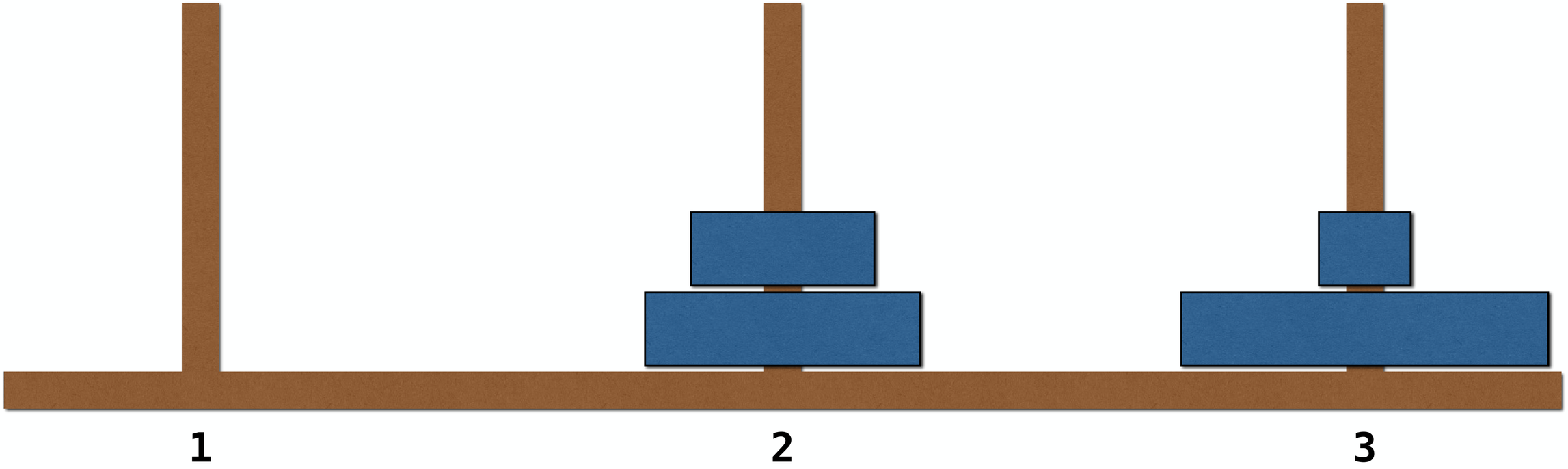

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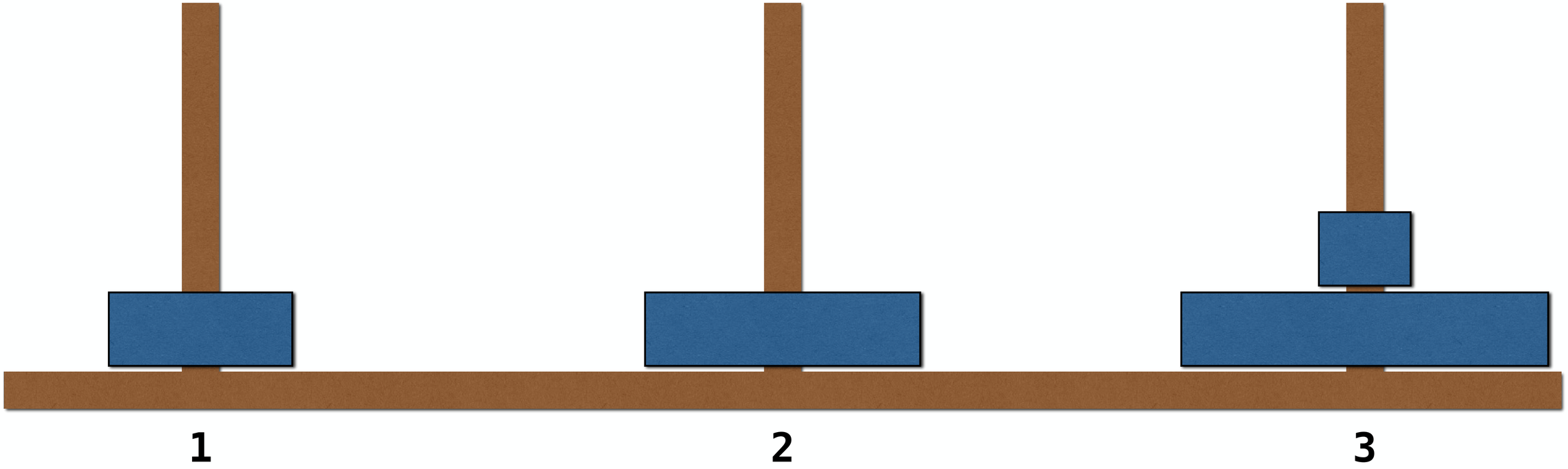

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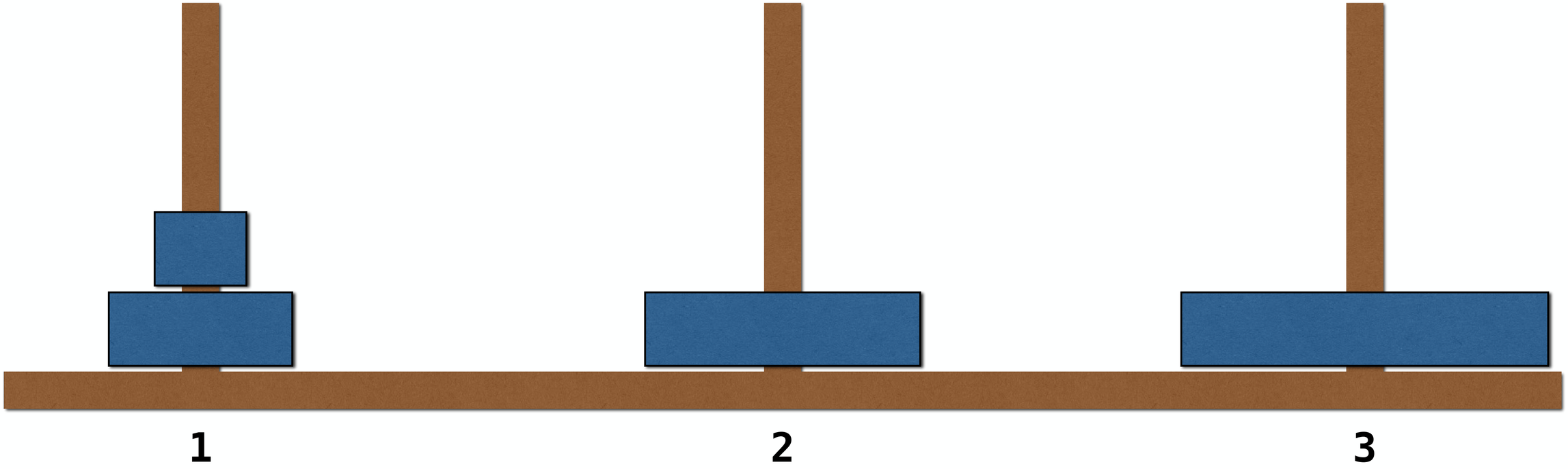

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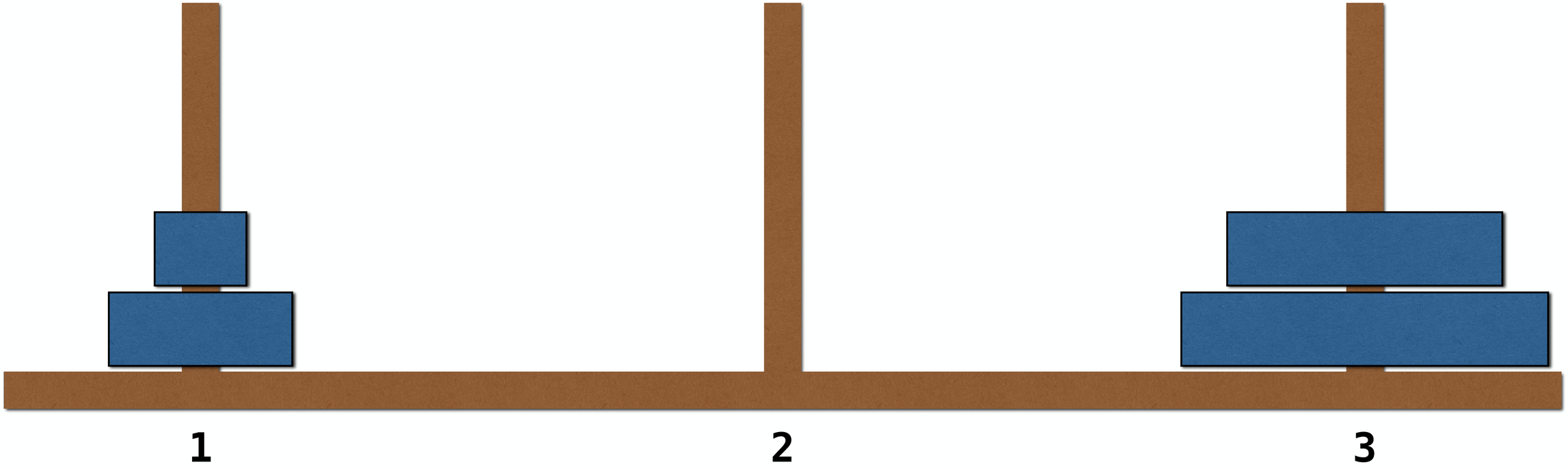

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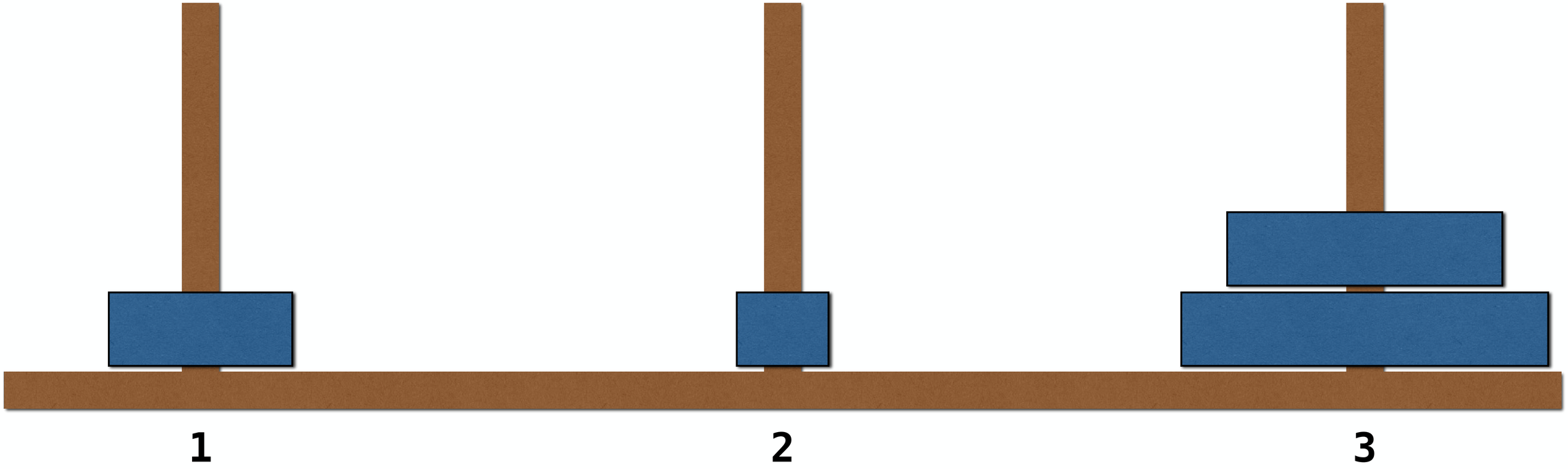

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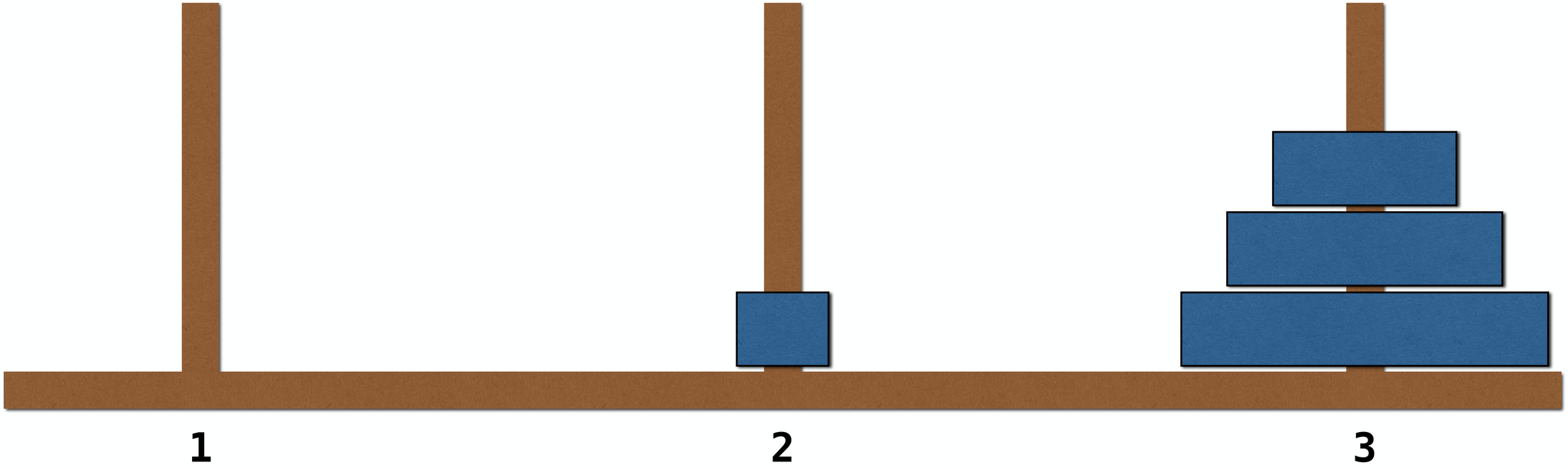

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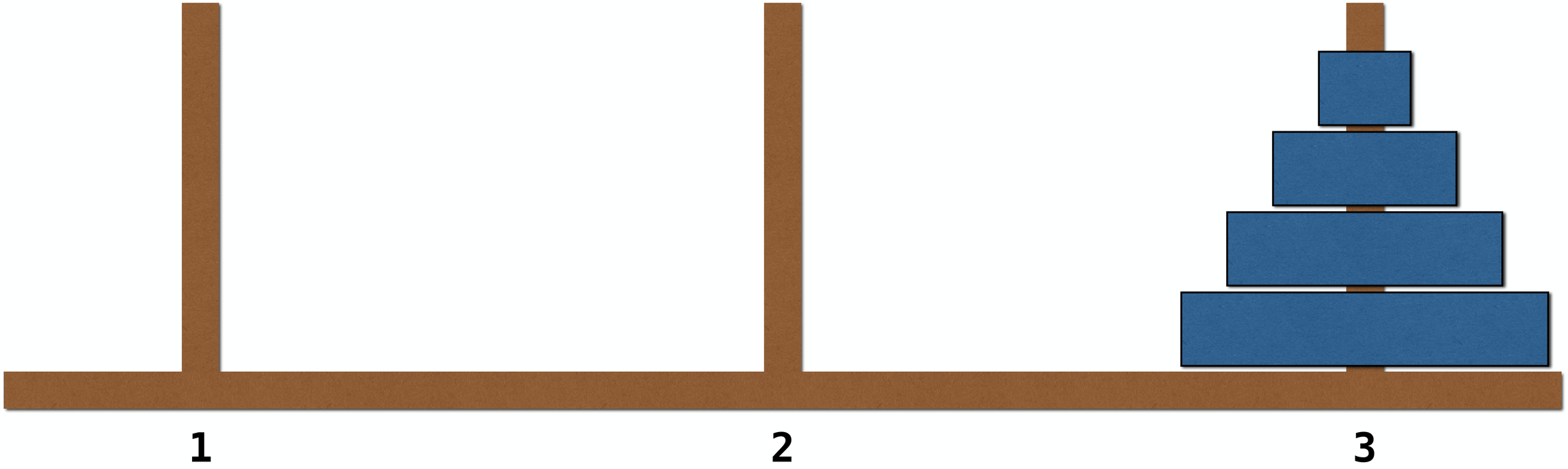
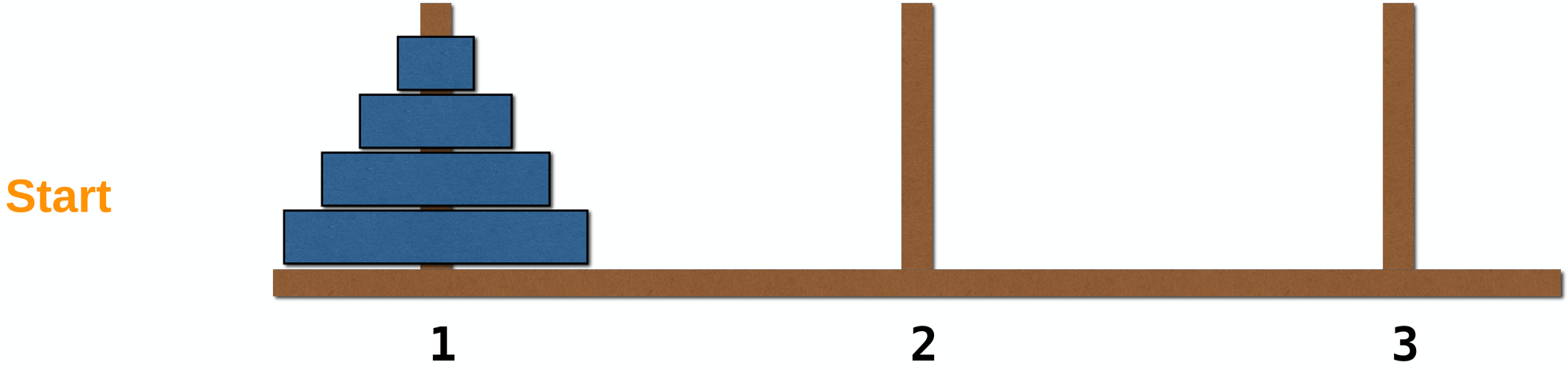


Idea

- Moving a tower of height `h = 0` is simple: nothing to do
- Moving a tower of height `h > 0` from position `a` to position `b` is decomposed into three subtasks
  1. Move the top `h - 1` slice to the free place (using recursion)
  2. Move the lowest slice to the target position
  3. Move the intermediate slices to the target position (using recursion)
- Moving a tower of height `h > 0` thus requires moving a tower of height `h - 1` twice



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- Moving a tower of height `h = 0` is simple: nothing to do
- Moving a tower of height `h > 0` from position `a` to position `b` is decomposed into three subtasks
  1. **Move the top `h - 1` slice to the free place (using recursion)**
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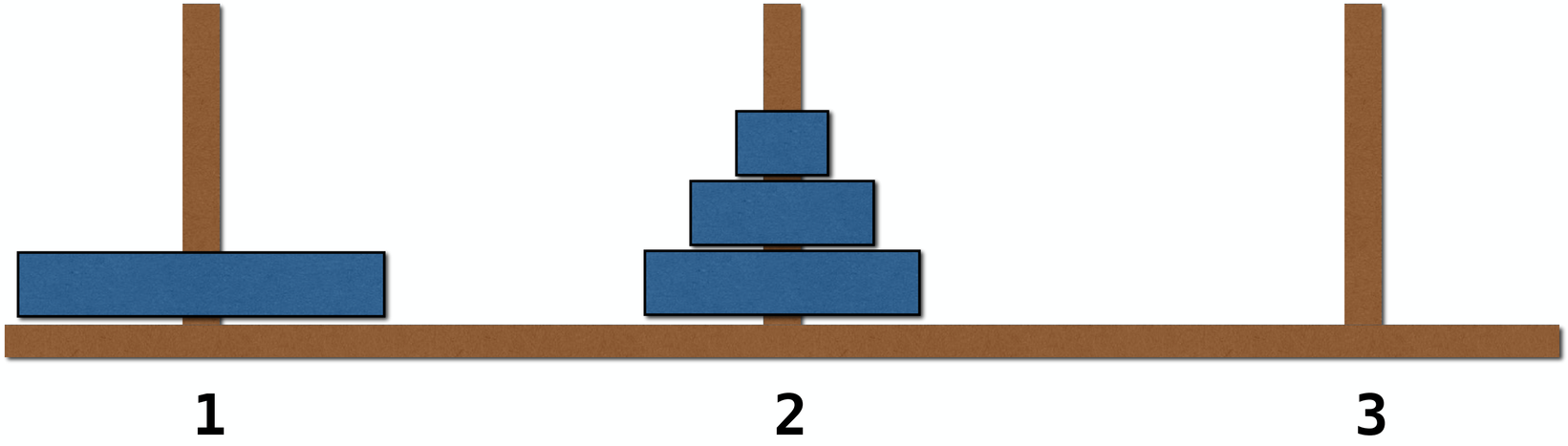
After step 1



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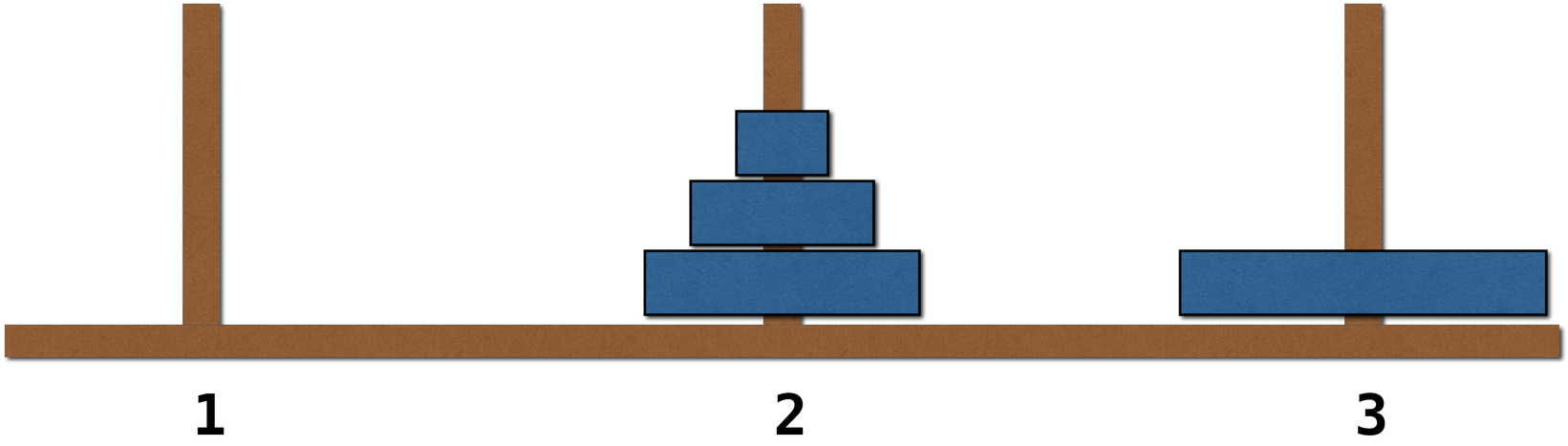
  2. **Move the lowest slice to the target position**
  3. Move the intermediate slices to the target position (using recursion)
- Moving a tower of height `h > 0` thus requires moving a tower of height `h - 1` twice

After step 2



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  1. Move the top `h - 1` slice to the free place (using recursion)
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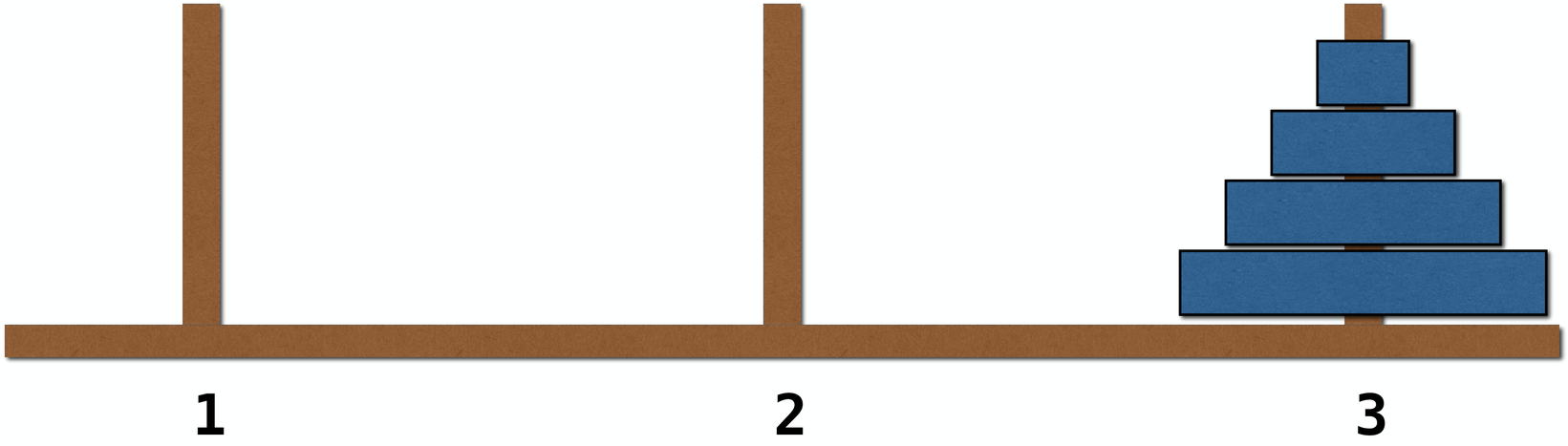
  3. **Move the intermediate slices to the target position (using recursion)**
- Moving a tower of height `h > 0` thus requires moving a tower of height `h - 1` twice

After step 3



Code

Input: spot 1, 2, or 3 where **a** and **b** must be different

Direct recursive method calls

```
public static void move(int height, int a, int b) {
    if (height > 0) {
        int c = free(a, b);      // c is the free spot
        move(height - 1, a, c); // move height - 1 slices from spot a to spot c
        // then the remaining slice from a to b
        System.out.println("move " + a + " to " + b);
        move(height - 1, c, b); // move height - 1 slices from spot c to spot b
    }
}
```

Remains to determine the free space ...

Free place

	1	2	3
1		3	2
2	3		1
3	2	1	

(read: **`free(x,y)`** is the position which is not occupied)

Apparently, the result depends only on the **sum** of the two arguments ...

	1	2	3
1		3	4
2	3		5
3	4	5	

Implementation

- We implement the method `free(...)` with `switch`

```
public static int free(int a, int b) {
    switch (a + b) {
        case 3:  return 3;
        case 4:  return 2;
        case 5:  return 1;
        default: return -1;
    }
}
```

Shorter alternative (more difficult to understand)

```
public static int free(int a, int b) {
    return 6 - (a + b);
}
```

Tower of Hanoi

```java
public class Hanoi {
    public static void move(int height, int a, int b) {
        if (height > 0) {
            int c = free(a, b);
            move(height - 1, a, c);
            System.out.println("move " + a + " to " + b);
            move(height - 1, c, b);
        }
    }

    public static int free(int a, int b) {
        return 6 - (a + b);
    }

    public static void main(String[] args) {
        int h = InputReader.readInt("Height of the tower: ");
        System.out.println("");
        move (h, 1, 3);
        System.out.println("");
    }
}
```

W12E02 - Recursion sum

Start exercise

Easy

Not started yet.

Due by tonight

10 min

3 pts

- **Problem statement**
  - Create a recursive function that determines the sum of integers **`from 1 to the value n`** (input parameter)
  - Implement the same function using iteration (e.g. using a for loop)

Outline

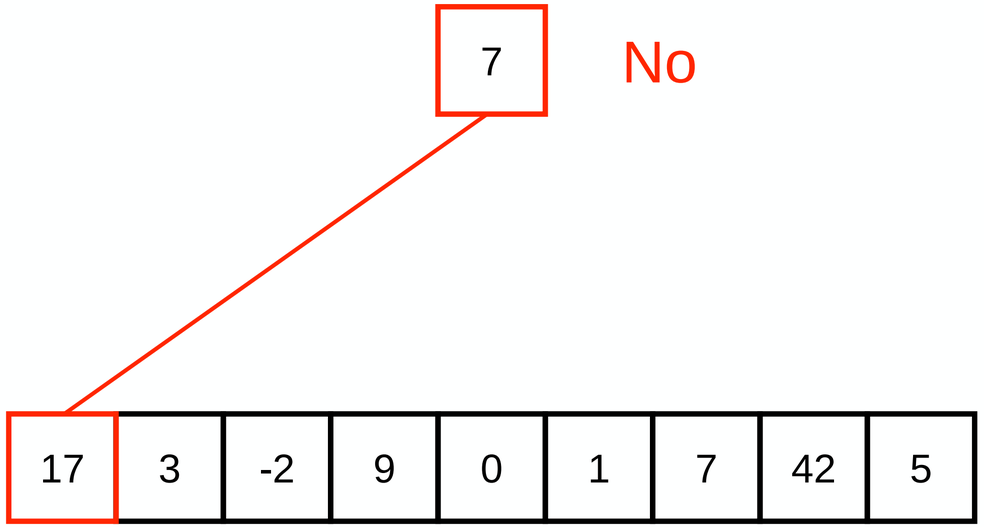
- The principle of recursion
- Example: Towers of Hanoi
- Example: Binary search
- Example: Koch snowflake
- Example: Mergesort

Binary search

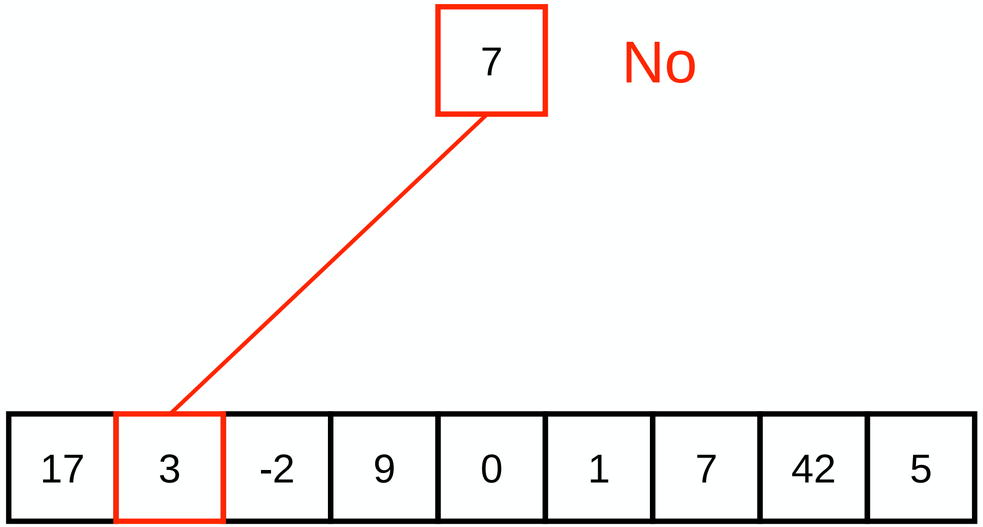
- Suppose we want to find out whether the element **`x = 7`** is contained in an **`array`**
- Naive approach
  - Compare **`7`** in order with the elements **`array[0], array[1],`** etc.
  - If there is an **`index`** with **`array[index] == 7`**, output **`index`**
  - Otherwise, output **`-1`** which means “not found"

```
public static int find(int[] array, int x) {
    int index = 0;
    while (index < array.length && array[index] != x) {
        index++;
    }
    if (index == array.length) {
        return -1;
    }
    else {
        return index;
    }
}
```

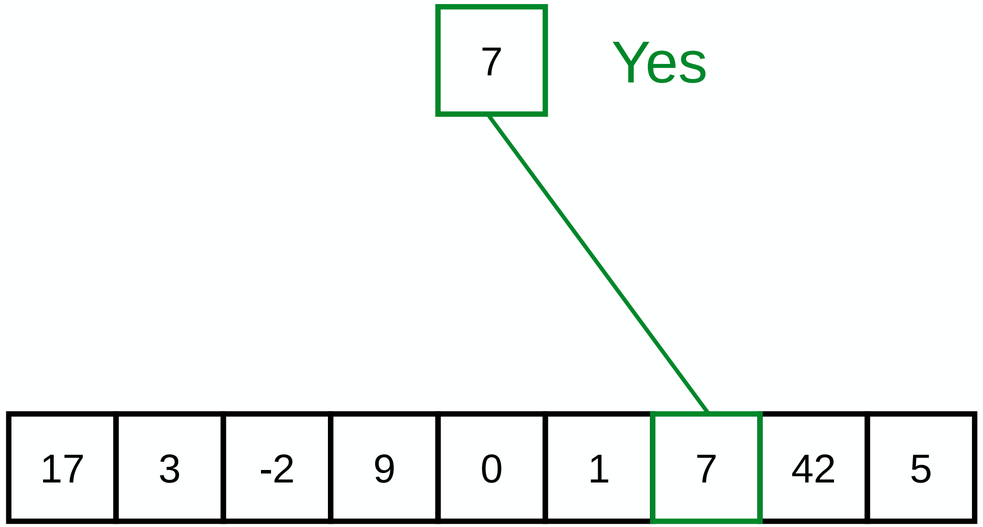
Naive search



Naive search



Naive search



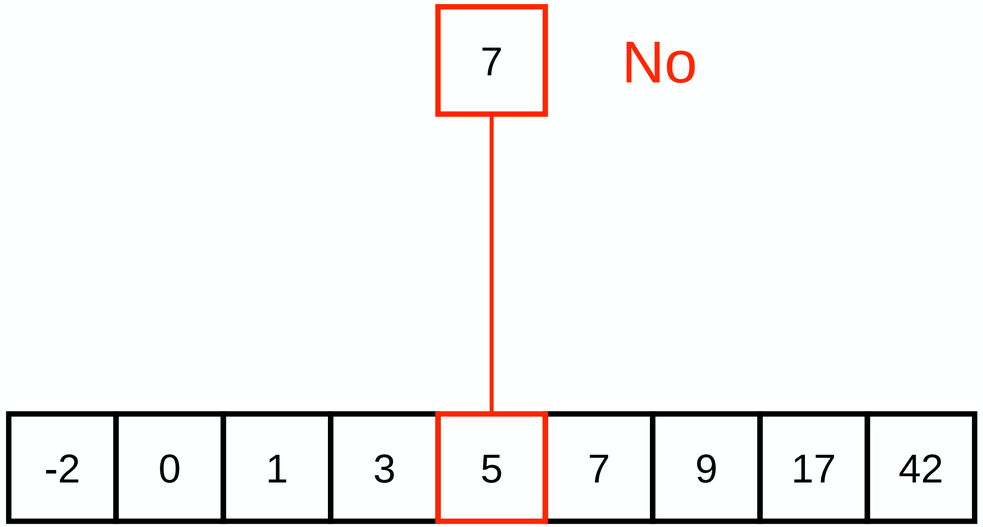
- Analysis: in the example, `7` comparisons are needed
- In the worst case, `n` comparisons are necessary for a field of length `n`
- If `7` actually occurs in the field, we need on average `(n + 1)/2` many comparisons
- Can we do it better?

Idea

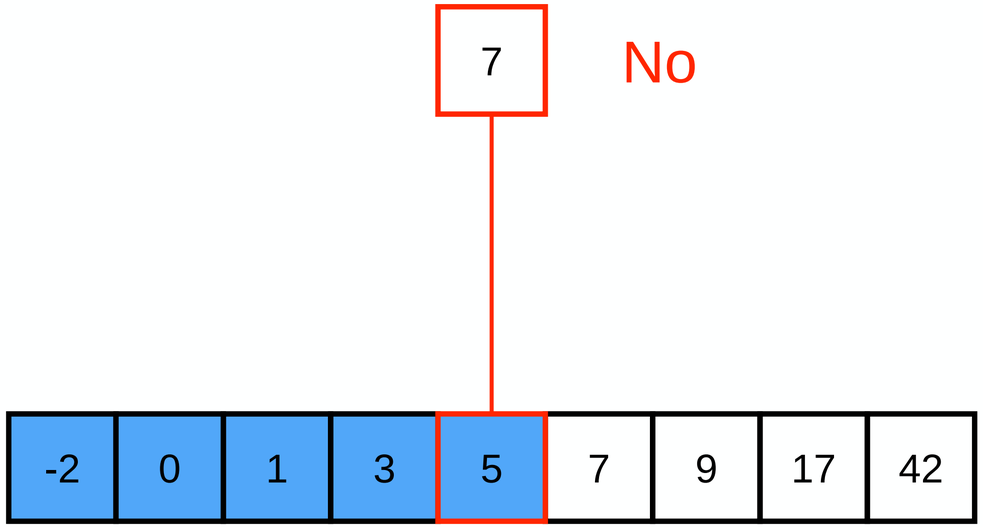
- Sort the field
- Compare **7** with the value in the middle
- if it's equal or if its elements match, the search is complete
- If **7** is smaller, we only need to continue searching on the left
- If **7** is greater, we only need to continue searching on the right

→ **Binary** search

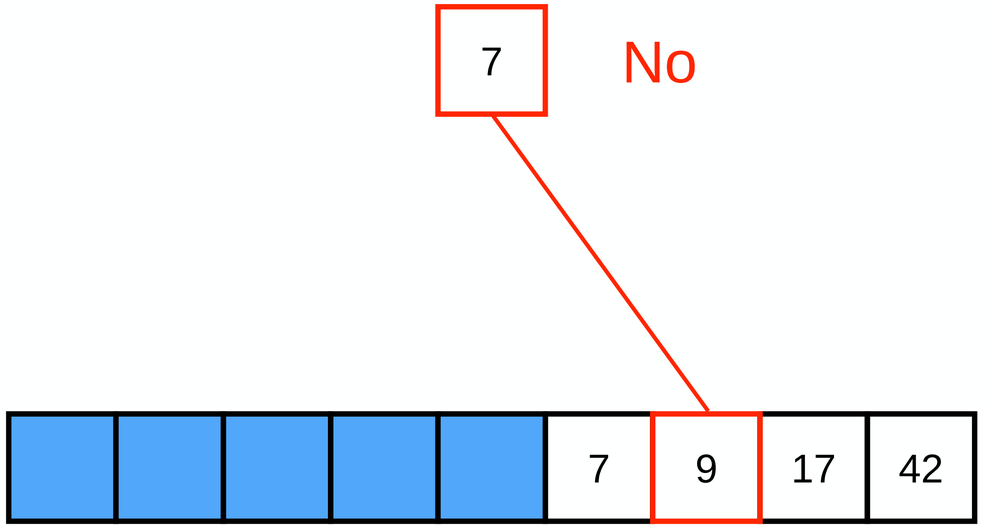
Binary search



Binary search



Binary search



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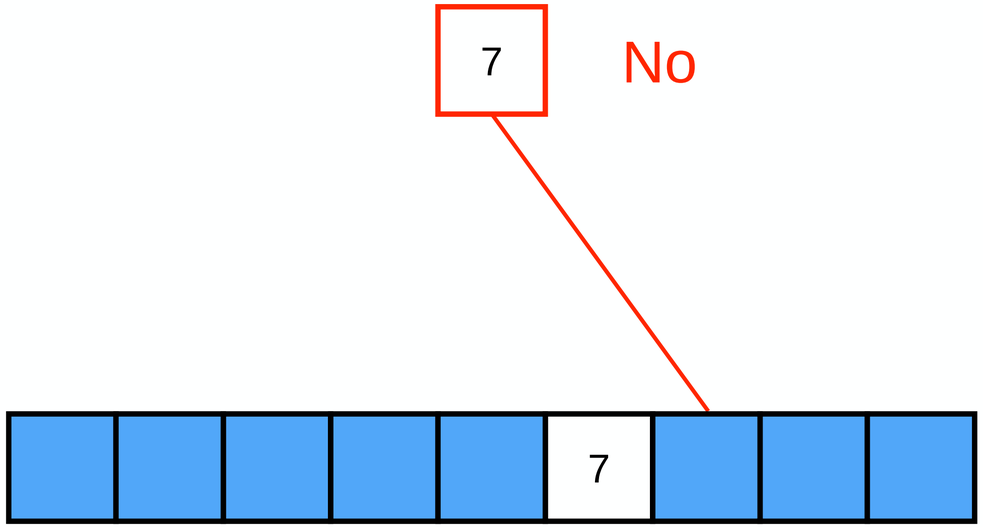
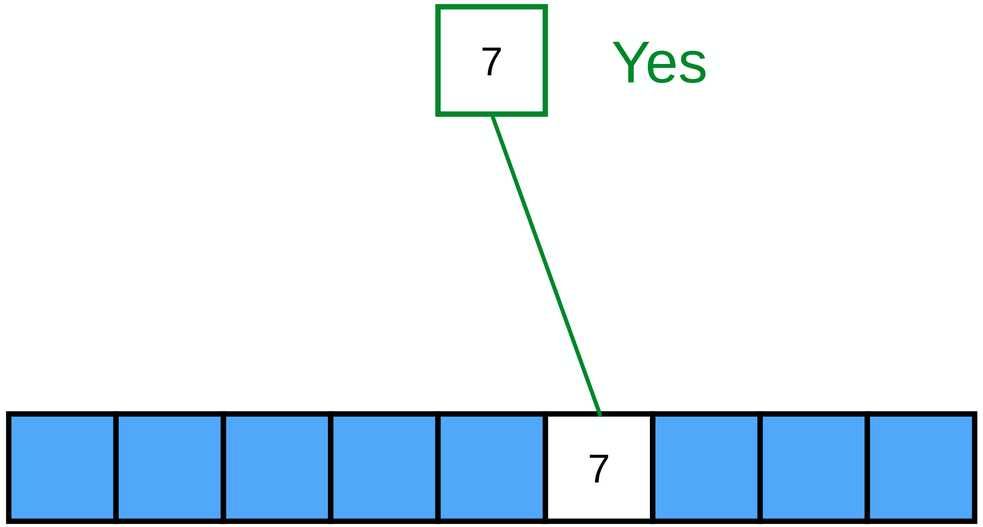


Binary search



- Analysis: binary search only needs **3** comparisons
- If the sorted array has $\mathbf{2^n-1}$ elements, at most **n** comparisons are necessary

Idea

- We introduce an auxiliary function that searches in the interval `[startIndex, endIndex]`

```
public static int find(int[] array, int element, int startIndex, int endIndex)
```

Then, we can invoke this method

```
public static int find(int[] array, int element) {
    return find(array, element, 0, array.length - 1);
}
```

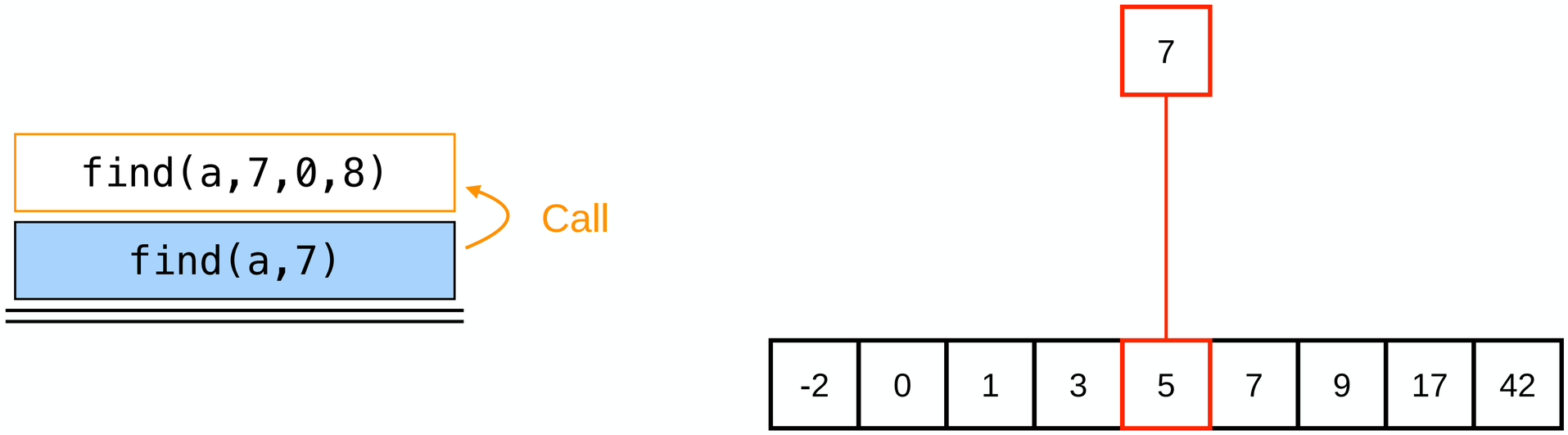
Implementation of binary search

```
public static int find(int[] array, int element, int startIndex, int endIndex) {
    int middle = (startIndex + endIndex) / 2;
    if (array[middle] == element) {
        return middle; // found
    }
    else if (startIndex == endIndex) {
        return -1; // not found
    }
    else if (element > array[middle]) {
        return find(array, element, middle + 1, endIndex);
    } else if (startIndex < middle) {
        return find(array, element, startIndex, middle - 1);
    } else {
        return -1; // not found
    }
}
```

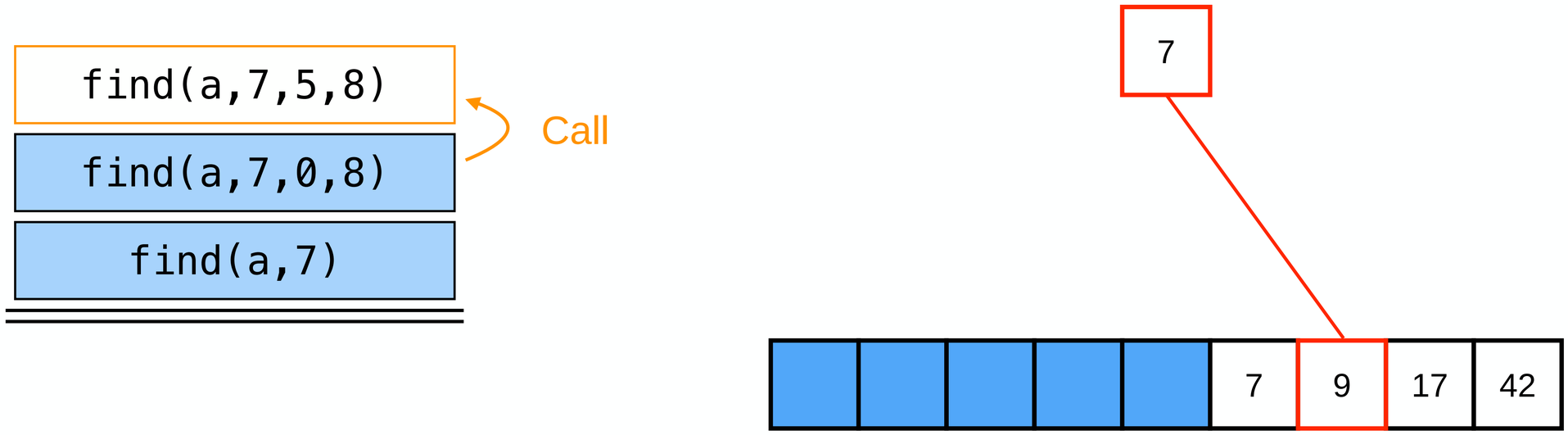
Execution

```
find(a,7)
```

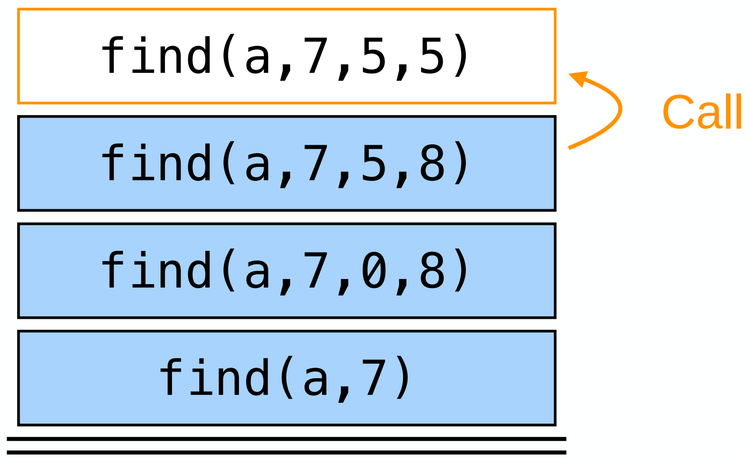
Execution

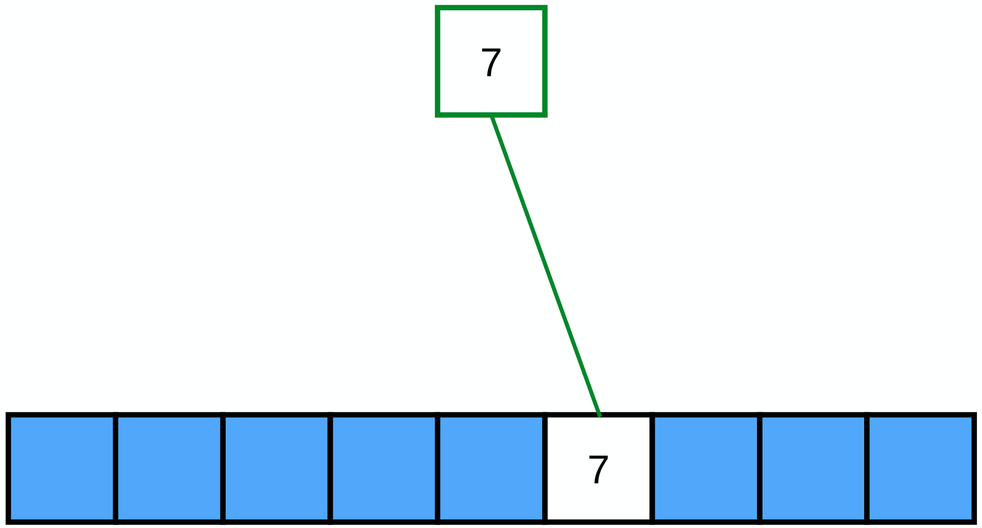


Execution

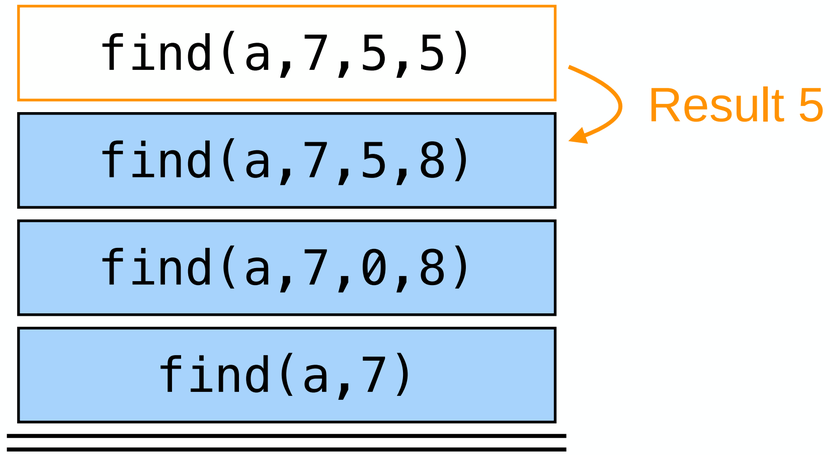


Execution

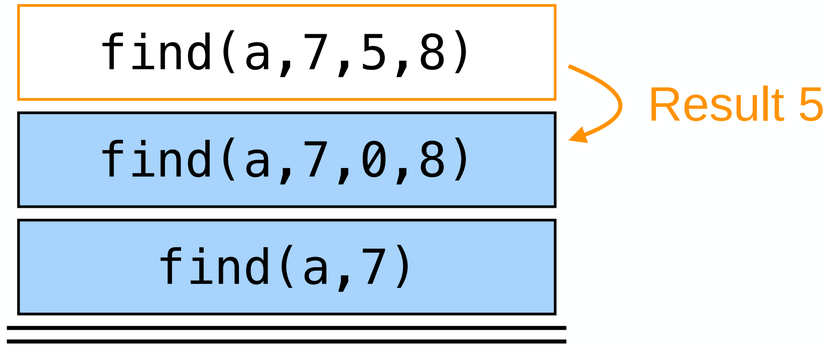




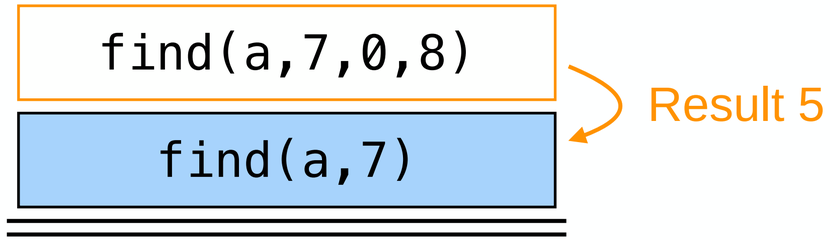
Execution



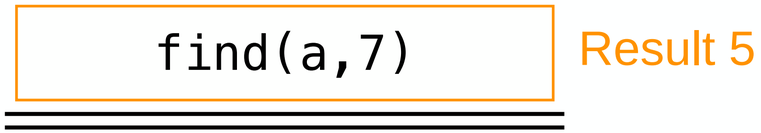
Execution



Execution



Execution



Observation

- The function calls are managed according to the LIFO principle (last in first out)

→ Data structure: **`stack`**

- Only the top/last call is active
- Note: there can be several further inactive calls of the same function at once

Termination

- To **prove** that `find()` terminates, we observe:
  1. If `startIndex` and `endIndex` are the same, `find()` is called for an interval `[n, n]` with **only one** element → the function call terminates directly
  2. If `find()` is called for an interval `[n1, n2]` with more than one element, then
     1. the call either terminates directly (because `element` was found), or
     2. `find()` is called with an interval that is genuinely contained in `[n1, n2]`, or more precisely: even contains at most half of the elements of `[n1, n2]`

→ Similar techniques are also used for other recursive functions

Iterative binary search

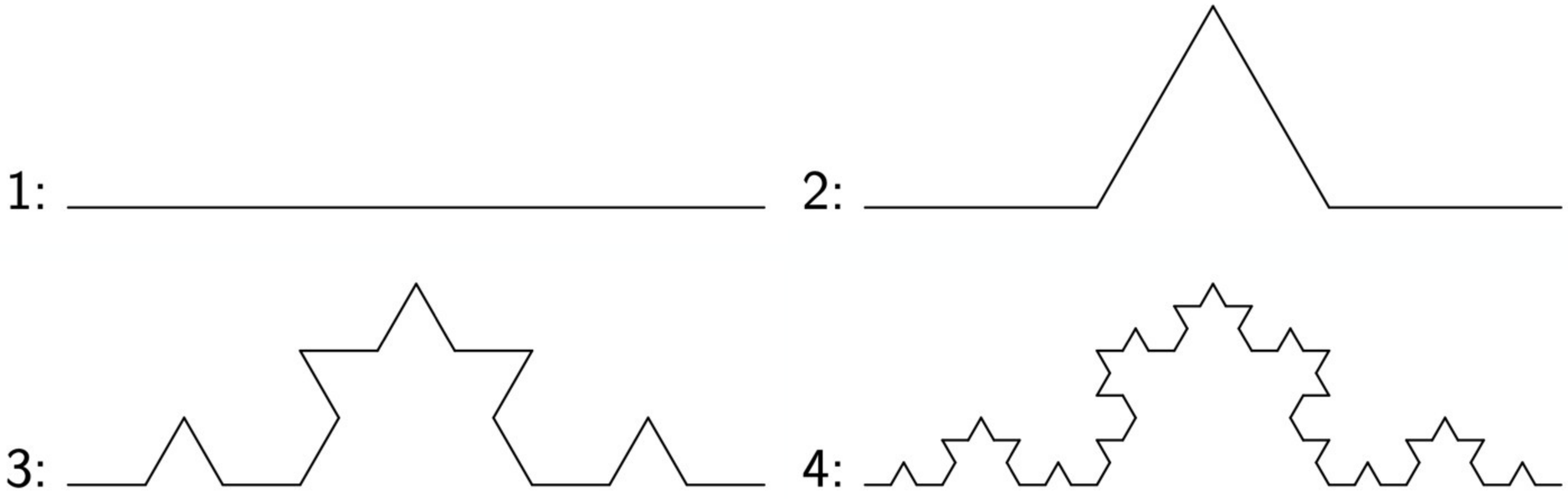
```java
public static int findIterative(int[] array, int element) {
    int startIndex = 0;
    int endIndex = array.length - 1;
    while (true) {
        int middle = (endIndex + startIndex) / 2;
        if (element == array[middle]) {
            return middle;
        } else if (startIndex == endIndex) {
            return -1; // not found
        } else if (element > array[middle]) {
            startIndex = middle + 1;
        } else if (startIndex < middle) {
            endIndex = middle - 1;
        } else {
            return -1; // not found
        }
    }
}
```

Outline

- The principle of recursion
- Example: Towers of Hanoi
- Example: Binary search
- Example: Koch snowflake
- Example: Mergesort

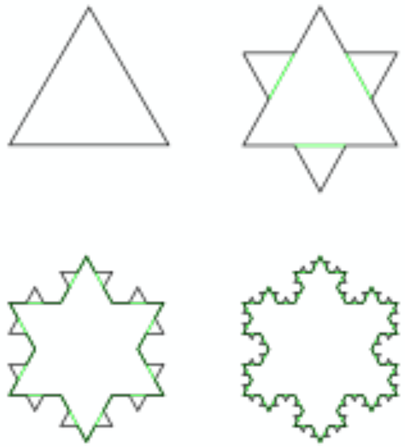
The Koch snowflake curve

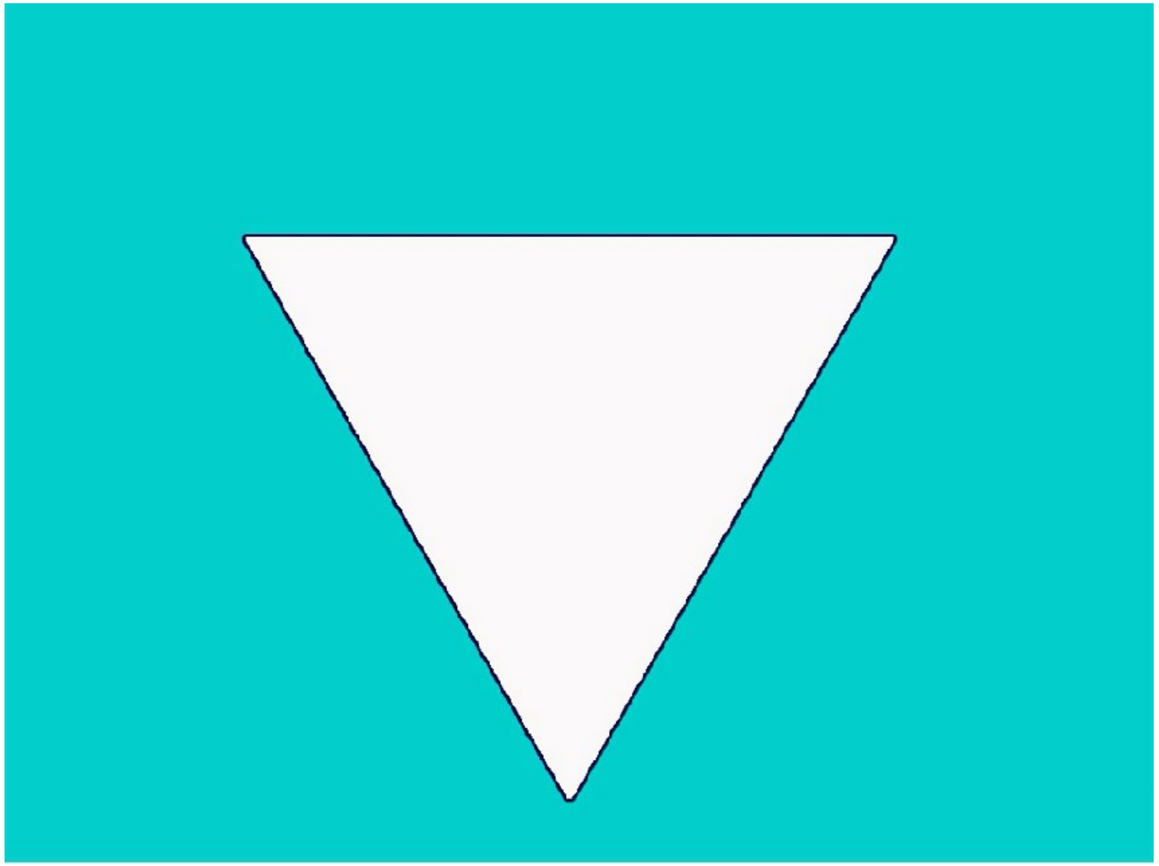
- The Koch curve is a fractal
- It is formed in the following steps
  - Start with a straight line
  - Remove the middle third from each straight line piece
  - Add an equilateral triangle instead



Flake math

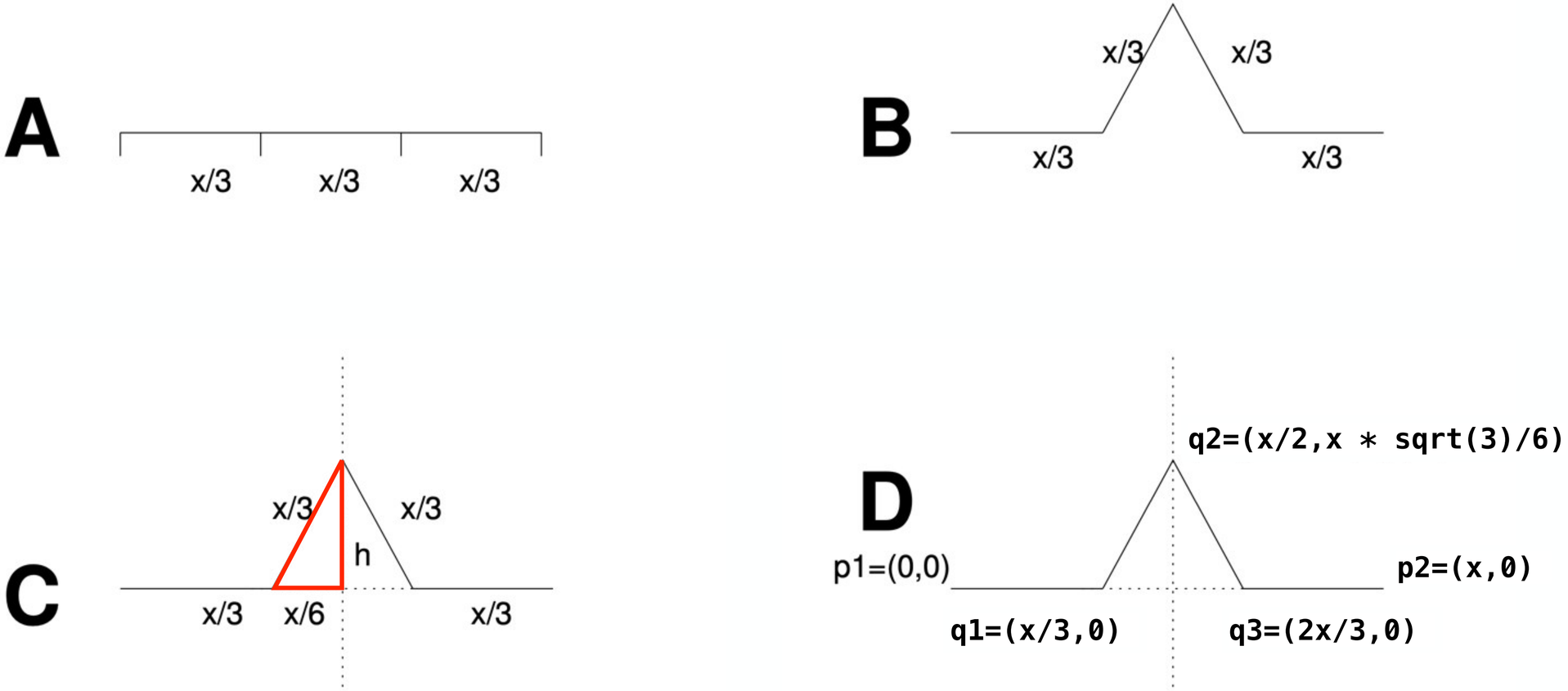
- If you form a Koch curve from each of the three sides of an equilateral triangle, you get the Koch snowflake





Flake formation: horizontal baseline

- From each straight line piece, remove the middle third and put on an equilateral triangle for it



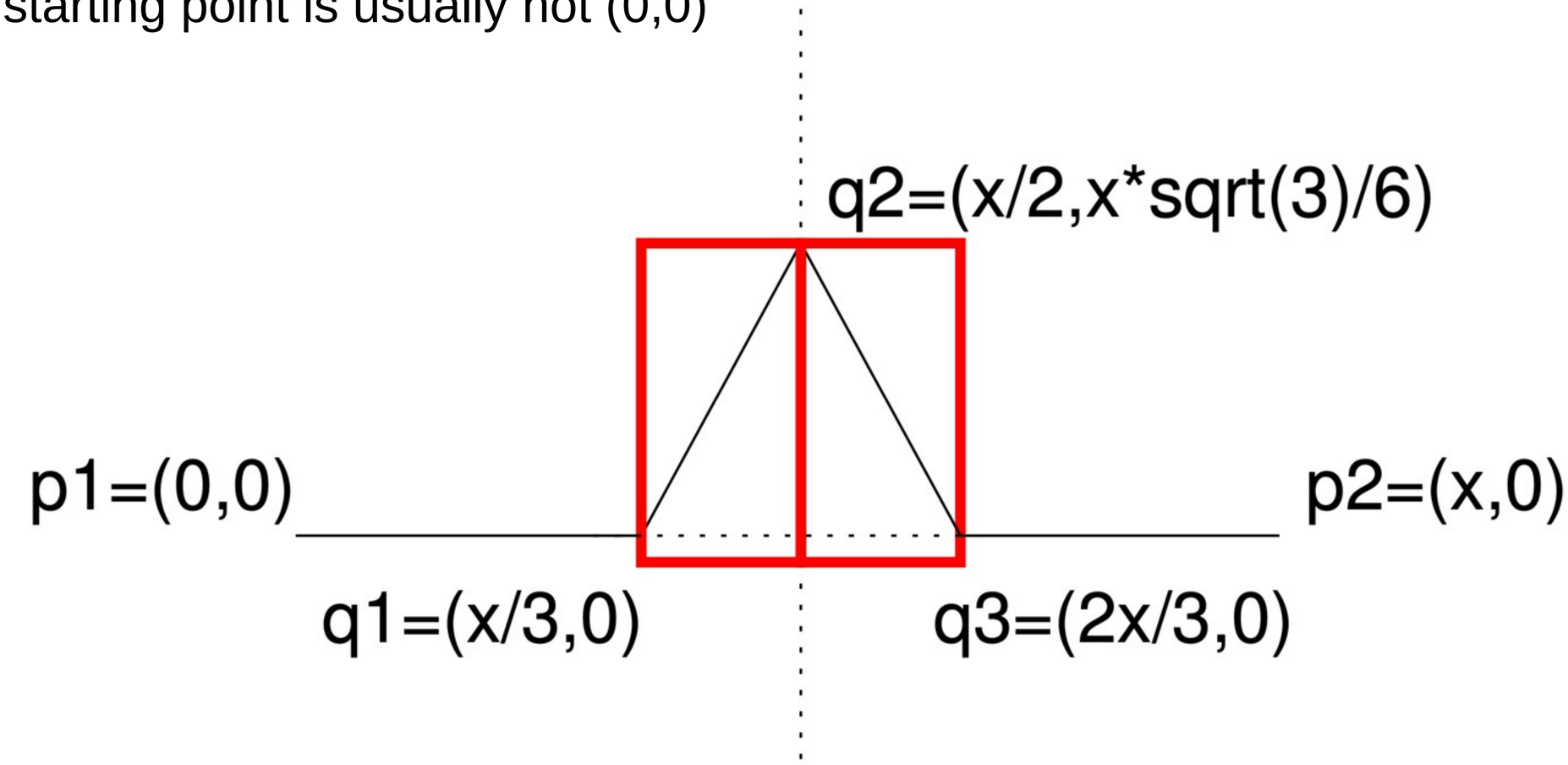
Pythagorean theorem for right triangle

$a^2 + b^2 = c^2$ $\quad$ $x^2/36 + h^2 = x^2/9$

$(x/6)^2 + h^2 = (x/3)^2$ $\quad$ $h = x * \mathrm{sqrt}(3)/6$

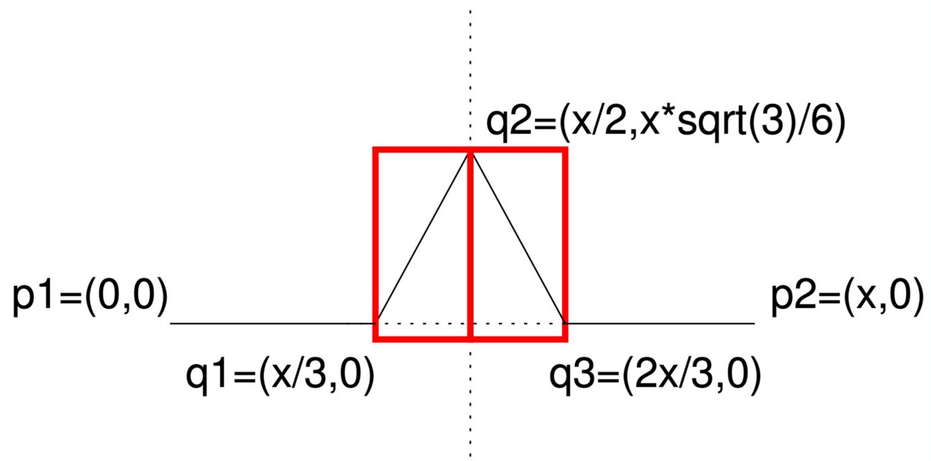
Flake formation: any baseline

- Procedure is applied recursively
  1. The baseline may not be horizontal and
  2. The starting point is usually not (0,0)



Solution: translation and rotation

- Use the simple calculation of the point coordinates
  1. Move the line to the origin by adding the negative coordinates of the first point, i.e. of **`(-x1,-y1)`**
  2. Rotate the shifted line by an angle ϕ so that it is horizontal
  3. Make the simple calculation
  4. Rotate the newly generated part of the flake in the opposite direction using **`-φ`** and shift it back by adding **`(x1, y1)`**



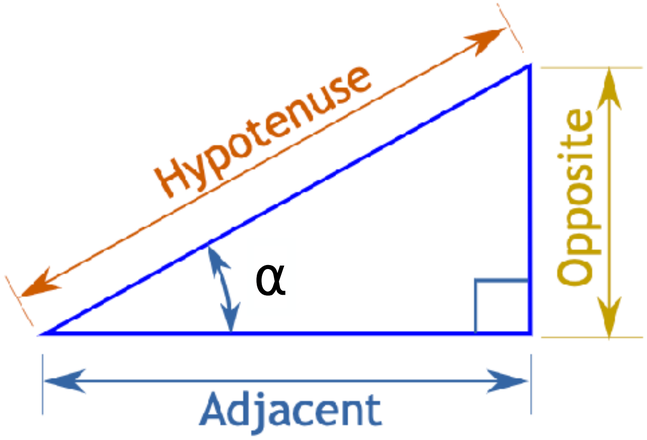
Basic math

- Sine, Cosine and Tangent (often shortened to sin, cos and tan) are each a ratio of sides of a right angled triangle

$$\sin \alpha = \frac{\text{Opposite}}{\text{Hypotenuse}}$$

$$\cos \alpha = \frac{\text{Adjacent}}{\text{Hypotenuse}}$$

$$\tan \alpha = \frac{\text{Opposite}}{\text{Adjacent}}$$

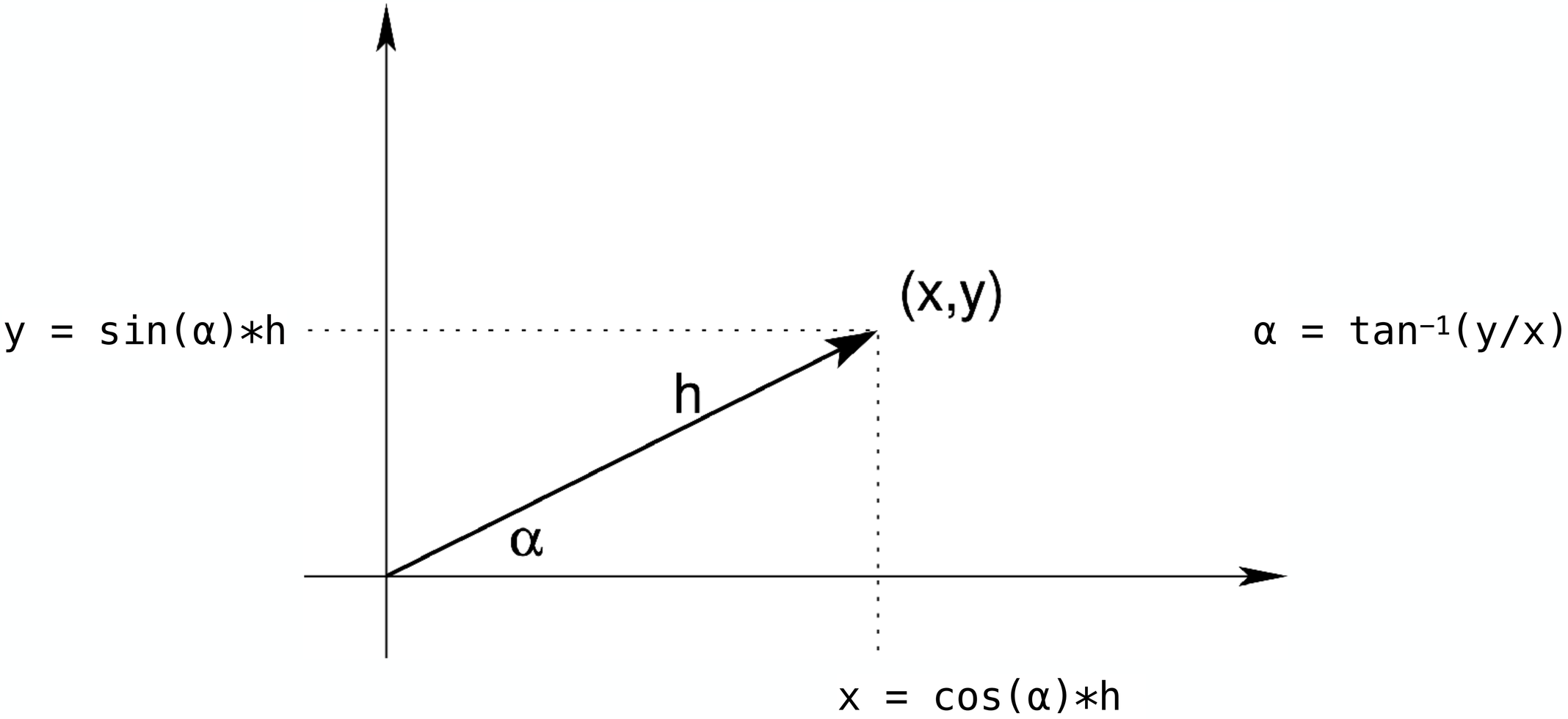


$$\alpha = \tan^{-1}\left(\frac{\text{Opposite}}{\text{Adjacent}}\right)$$

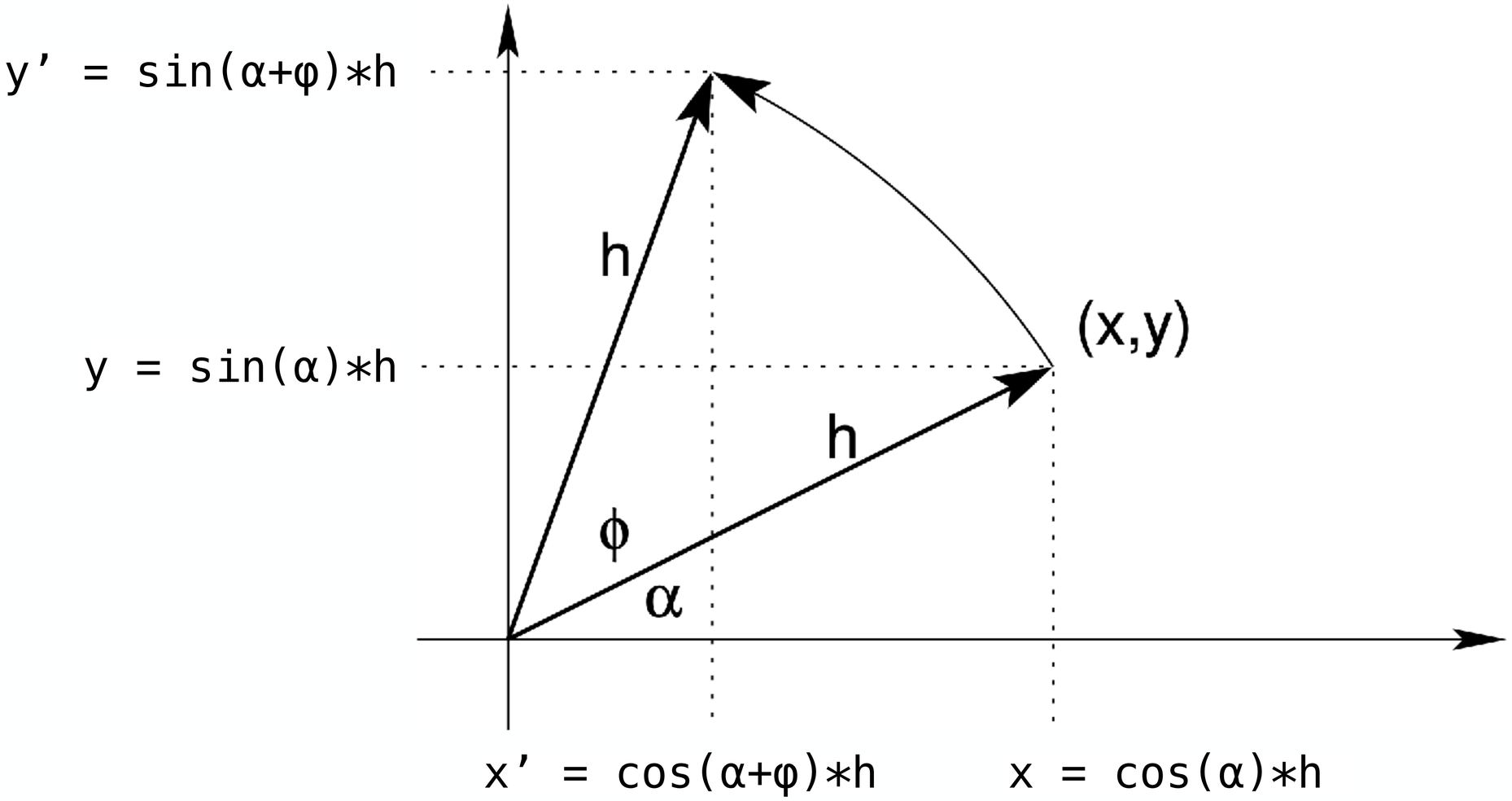
also called `arctan` or **`atan`**

https://www.mathsisfun.com/sine-cosine-tangent.html

Flake math: rotation

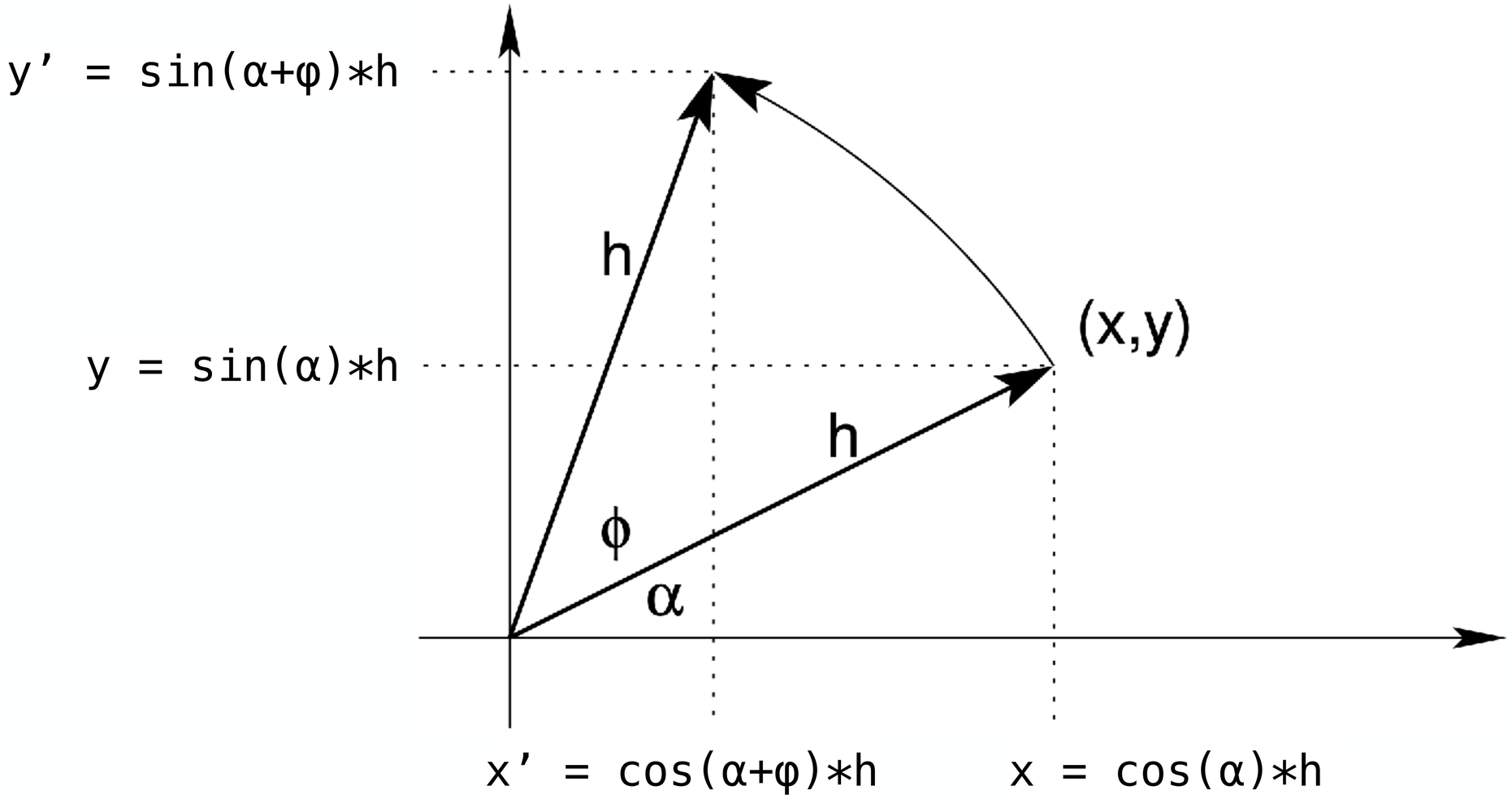


Flake math: rotation



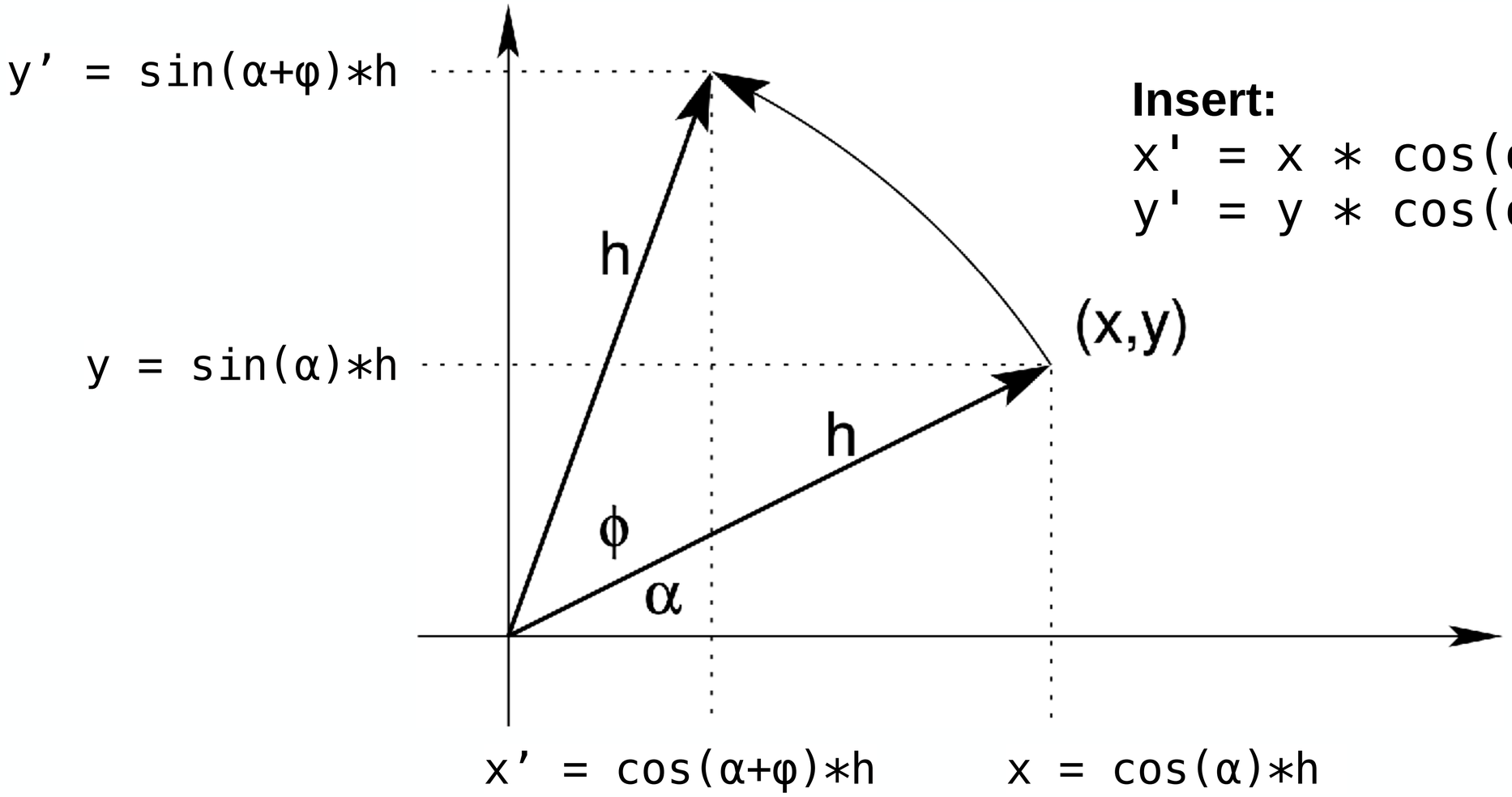
Flake math: rotation

- Addition theorem 1: `sin(α + φ) = sin(α) ∗ cos(φ) + sin(φ) ∗ cos(α)`
- Addition theorem 2: `cos(α + φ) = cos(α) ∗ cos(φ) − sin(φ) ∗ sin(α)`



Flake math: rotation

- Addition theorem 1: `sin(α + φ) = sin(α) ∗ cos(φ) + sin(φ) ∗ cos(α)`
- Addition theorem 2: `cos(α + φ) = cos(α) ∗ cos(φ) − sin(φ) ∗ sin(α)`



Insert:

```
x' = x ∗ cos(φ) − y ∗ sin(φ)
y' = y ∗ cos(φ) + x ∗ sin(φ)
```

Intermediate calculation

- `x' = cos(α + φ)  * h`
- `x' = (cos(α) * cos(φ) – sin(φ) * sin(α)) * h`

Substitute sin(α) = y / h
Substitute cos(α) = x / h

- `x' = (x / h * cos(φ) – sin(φ) * y / h)  * h`
- **`x' =  x * cos(φ)  –  y * sin(φ)`**

- `y' =  sin(α + φ)  * h`
- `y' = (sin(α) * cos(φ) + sin(φ) * cos(α)) * h`

Substitute sin(α) = y / h
Substitute cos(α) = x / h

- `y' = (y / h * cos(φ) + sin(φ) * x / h)  * h`
- **`y' =  y * cos(φ)  +  x * sin(φ)`**

Realization in Java

- Use the class **`Point`**
- Realize the iteration by recursion over the single partial routes
  - Base case: draw the current route
  - Recursion case: calculate the intermediate points of the current path and iterate over all the resulting partial distances
- Calculation of the intermediate points with the help of the operations from the class **`Point`**

Point

```java
public class Point {
    private double x, y;
    Point(double x, double y) { set(x, y); }
    public double getX() { return x; }
    public double getY() { return y; }
    private void set(double x, double y) {
        this.x = x;
        this.y = y;
    }

    Point rotate(double phi) {
        double newX = x * Math.cos(phi) - y * Math.sin(phi);
        double newY = y * Math.cos(phi) + x * Math.sin(phi);
        set(newX, newY);
        return this;
    }

    Point translate(double x, double y) {
        set(this.x + x, this.y + y);
        return this;
    }

    @Override
    public String toString() {
        return "(" + x + "," + y + ')';
    }
}
```

Changes the object on which the method is executed and returns this changed object

Changes the object on which the method is executed and returns this changed object

Flake

```java
import java.util.*;

public class Flake {

    private final List<Point> points = new ArrayList<>();

    Flake(Point p1, Point p2) {
        points.add(p1);
        points.add(p2);
    }

    public List<Point> getPoints() {
        return points;
    }

    //...

    @Override
    public String toString() {
        return "Flake{" + points + '}';
    }
}
```

Flake.recursiveSnow

```java
public void recursiveSnow(Point startPoint, Point endPoint, int depth) {
    if (depth > 0) {
        double shiftX = startPoint.getX(), shiftY = startPoint.getY();
        endPoint.translate(-shiftX, -shiftY);

        double angle = Math.atan2(endPoint.getY(), endPoint.getX());
        endPoint.rotate(-angle);

        Point helperPoint = new Point(endPoint.getX() / 3.0, 0.0);
        Point mid1 = helperPoint.rotate(angle).translate(shiftX, shiftY);
        helperPoint = new Point(endPoint.getX() / 2.0, endPoint.getX() / 6.0 * Math.sqrt(3.0));
        Point mid2 = helperPoint.rotate(angle).translate(shiftX, shiftY);
        helperPoint = new Point(2.0 * endPoint.getX() / 3.0, 0.0);
        Point mid3 = helperPoint.rotate(angle).translate(shiftX, shiftY);

        points.addAll(points.indexOf(startPoint) + 1, List.of(mid1, mid2, mid3));

        endPoint.rotate(angle).translate(shiftX, shiftY);

        recursiveSnow(startPoint, mid1, depth - 1);
        recursiveSnow(mid1, mid2, depth - 1);
        recursiveSnow(mid2, mid3, depth - 1);
        recursiveSnow(mid3, endPoint, depth - 1);
    }
}
```

Flake.recursiveSnow (with comments)

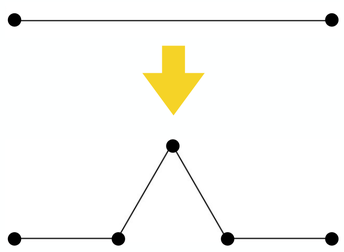
Important: terminating condition

```
public void recursiveSnow(Point startPoint, Point endPoint, int depth) {
    if (depth > 0) {
        double shiftX = startPoint.getX(), shiftY = startPoint.getY();
        endPoint.translate(-shiftX, -shiftY);

        double angle = Math.atan2(endPoint.getY(), endPoint.getX());
        endPoint.rotate(-angle);

        Point helperPoint = new Point(endPoint.getX() / 3.0, 0.0);
        Point mid1 = helperPoint.rotate(angle).translate(shiftX, shiftY);
        helperPoint = new Point(endPoint.getX() / 2.0, endPoint.getX() / 6.0 * Math.sqrt(3.0));
        Point mid2 = helperPoint.rotate(angle).translate(shiftX, shiftY);
        helperPoint = new Point(2.0 * endPoint.getX() / 3.0, 0.0);
        Point mid3 = helperPoint.rotate(angle).translate(shiftX, shiftY);

        points.addAll(points.indexOf(startPoint) + 1, List.of(mid1, mid2, mid3));

        endPoint.rotate(angle).translate(shiftX, shiftY);

        recursiveSnow(startPoint, mid1, depth - 1);
        recursiveSnow(mid1, mid2, depth - 1);
        recursiveSnow(mid2, mid3, depth - 1);
        recursiveSnow(mid3, endPoint, depth - 1);
    }
}
```

returns the angle of the point

Rotate end point to a horizontal line

Compute three new **mid points** using **rotate** and then **translate**

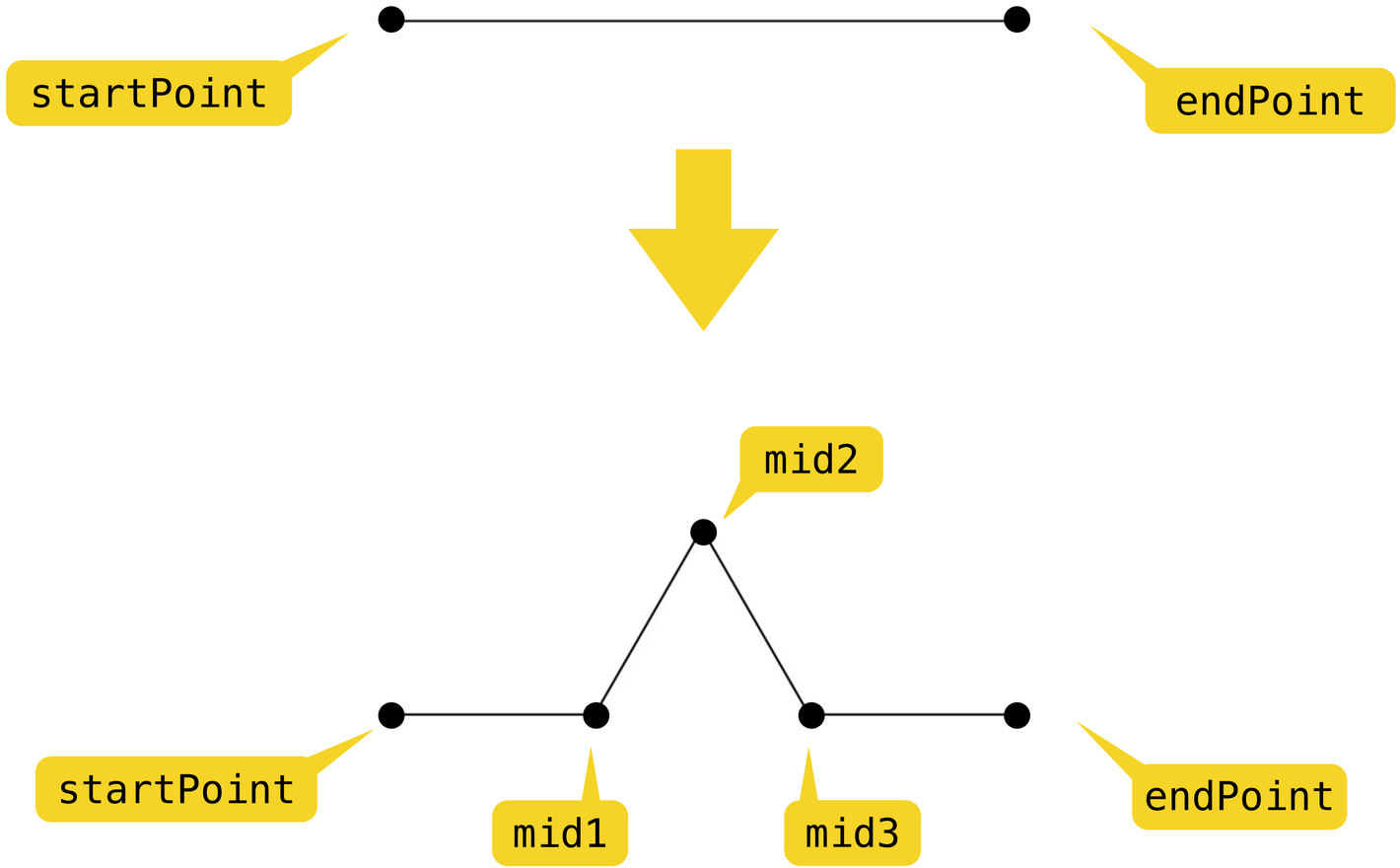


Create new polygon: add 3 mid points after start point

Undo translation and rotation

Recursion: create new flake for each segment

Calculation of mid points



Flake.snow (iterative)

```java
public void snow(int depth) {
    for (int i = 0; i < depth; i++) {
        List<Point> newPoints = new ArrayList<>();
        Point endPoint = null;
        for (int j = 0; j < points.size() - 1; j++) {
            Point startPoint = points.get(j);
            endPoint = points.get(j + 1);
            double shiftX = startPoint.getX(), shiftY = startPoint.getY();
            endPoint.translate(-shiftX, -shiftY);
            // rotate to horizontal
            double angle = Math.atan2(endPoint.getY(), endPoint.getX());
            endPoint.rotate(-angle);
            // compute three new mid points using rotate and then translate
            Point helperPoint = new Point(endPoint.getX() / 3.0, 0.0);
            Point mid1 = helperPoint.rotate(angle).translate(shiftX, shiftY);
            helperPoint = new Point(endPoint.getX() / 2.0, endPoint.getX() / 6.0 * Math.sqrt(3.0));
            Point mid2 = helperPoint.rotate(angle).translate(shiftX, shiftY);
            helperPoint = new Point(2.0 * endPoint.getX() / 3.0, 0.0);
            Point mid3 = helperPoint.rotate(angle).translate(shiftX, shiftY);
            // create new polygon: add 3 mid points after start point
            newPoints.addAll(List.of(startPoint, mid1, mid2, mid3));
            // undo translation and rotation
            endPoint.rotate(angle).translate(shiftX, shiftY);
        }
        newPoints.add(endPoint);
        points.clear();
        points.addAll(newPoints);
    }
}
```

Flake.snow (iterative, with comments)

```java
public void snow(int depth) {
    for (int i = 0; i < depth; i++) {
        List<Point> newPoints = new ArrayList<>();
        Point endPoint = null;
        for (int j = 0; j < points.size() - 1; j++) {
            Point startPoint = points.get(j);
            endPoint = points.get(j + 1);
            double shiftX = startPoint.getX(), shiftY = startPoint.getY();
            endPoint.translate(-shiftX, -shiftY);
            // rotate to horizontal
            double angle = Math.atan2(endPoint.getY(), endPoint.getX());
            endPoint.rotate(-angle);
            // compute three new mid points using rotate and then translate
            Point helperPoint = new Point(endPoint.getX() / 3.0, 0.0);
            Point mid1 = helperPoint.rotate(angle).translate(shiftX, shiftY);
            helperPoint = new Point(endPoint.getX() / 2.0, endPoint.getX() / 6.0 * Math.sqrt(3.0));
            Point mid2 = helperPoint.rotate(angle).translate(shiftX, shiftY);
            helperPoint = new Point(2.0 * endPoint.getX() / 3.0, 0.0);
            Point mid3 = helperPoint.rotate(angle).translate(shiftX, shiftY);
            // create new polygon: add 3 mid points after start point
            newPoints.addAll(List.of(startPoint, mid1, mid2, mid3));
            // undo translation and rotation
            endPoint.rotate(angle).translate(shiftX, shiftY);
        }
        newPoints.add(endPoint);
        points.clear();
        points.addAll(newPoints);
    }
}
```

Use iteration instead of recursion

Iterate through the array and get the current **`startPoint`** and **`endPoint`** for each segment

Do not add the **`endPoint`**, it will be processed and added in the next iteration (inner for loop)

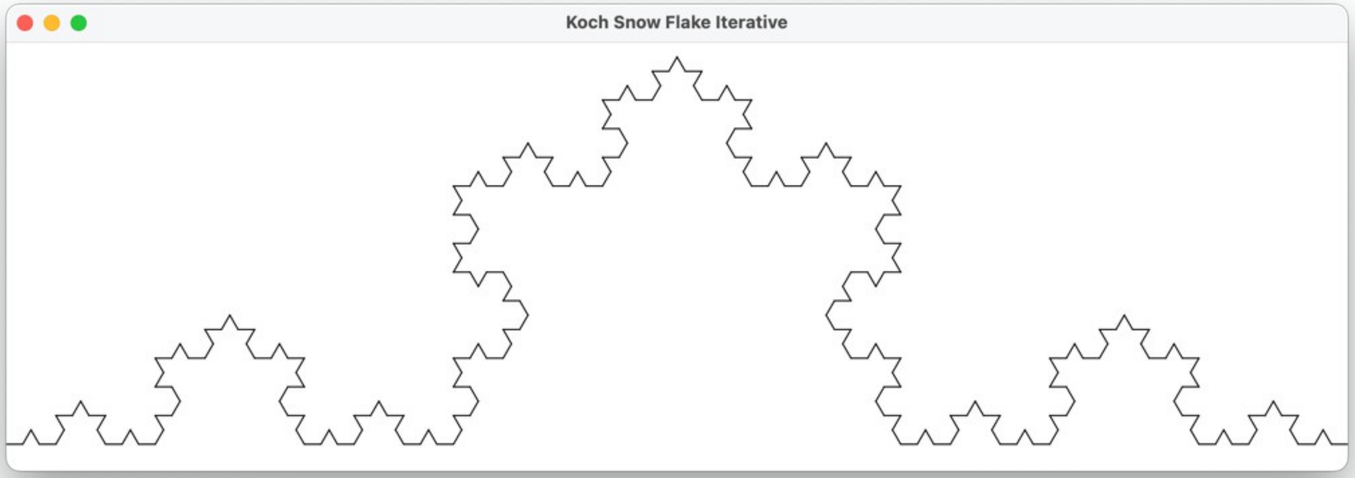
Add the **`endPoint`** (once for the whole iteration) and use all new points from now on

Usage

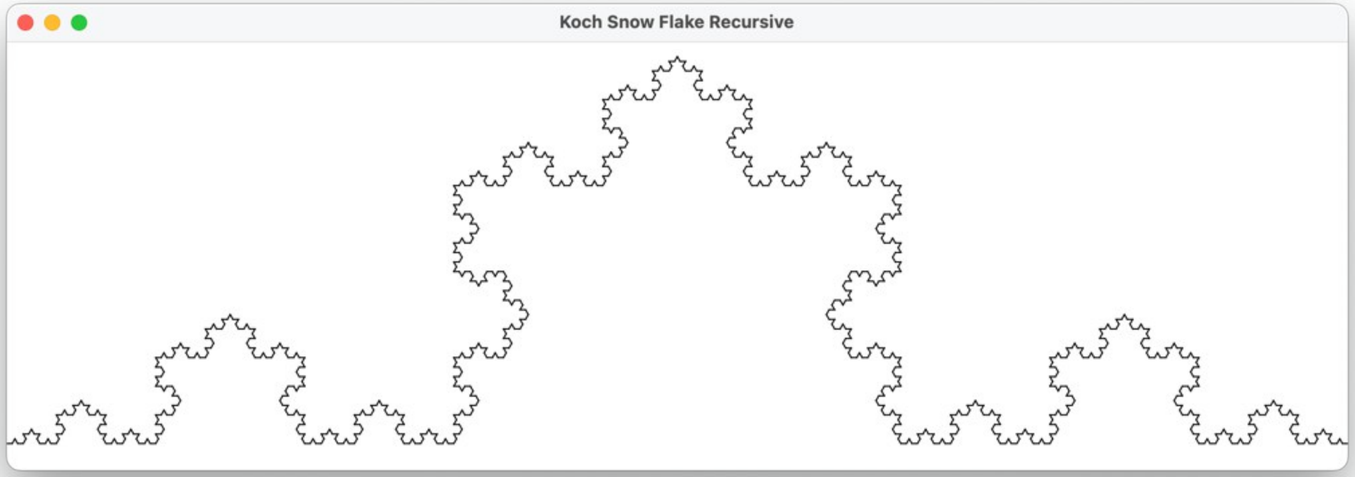
```java
package de.tum.in.ase;

import javafx.application.Application;
import javafx.scene.Scene;
import javafx.scene.layout.Pane;
import javafx.scene.shape.Line;
import javafx.stage.Stage;

public class KochSnowflake extends Application {

    @Override
    public void start(Stage primaryStage) {
        Point startPoint = new Point(0, 0);
        Point endPoint = new Point(1000, 0);

        Flake recursiveFlake = new Flake(startPoint, endPoint);
        recursiveFlake.recursiveSnow(startPoint, endPoint, 5);

        Flake iterativeFlake = new Flake(startPoint, endPoint);
        iterativeFlake.snow(4);

        Pane pane = new Pane();
        for (int i = 0; i < iterativeFlake.getPoints().size() - 1; i++) {
            Point p1 = iterativeFlake.getPoints().get(i);
            Point p2 = iterativeFlake.getPoints().get(i + 1);
            pane.getChildren().add(new Line(p1.getX(), 300 - p1.getY(), p2.getX(), 300 - p2.getY()));
        }
        primaryStage.setScene(new Scene(pane, 1000, 320));
        primaryStage.setTitle("Koch Snow Flake Iterative");
        primaryStage.show();
    }
}
```

Output

Depth: 4



Depth: 5



You can find the whole example on https://github.com/ls1intum/Snow-Flake-Exercise

W12E03 - N-th power of 10

Start exercise

Easy

Not started yet.

Due by tonight

5 min

2 pts

- **Problem statement**: Calculate the **`n-th power of 10`** using recursion
- Examples
  - `powerOf10(0) = 1`
  - `powerOf10(1) = 10`
  - `powerOf10(5) = 100.000`

Outline

- The principle of recursion
- Example: Towers of Hanoi
- Example: Binary search
- Example: Koch snowflake
- Example: Mergesort

Review: **mergesort** step 3 (iterative)

```java
public static <T extends Comparable<T>> List<T> merge(List<T> a, List<T> b) {
    List<T> result = new ArrayList<>();
    int i = 0;  // index for input list a
    int j = 0;  // index for input list b

    while (i < a.size() && j < b.size()) {
        if (a.get(i).compareTo(b.get(j)) < 0) {
            result.add(a.get(i++));
        } else {
            result.add(b.get(j++));
        }
    }

    while (i < a.size()) {
        result.add(a.get(i++));
    }
    while (j < b.size()) {
        result.add(b.get(j++));
    }

    return result;
}
```

Take the smaller element

Append all remaining elements of a into result

Append all remaining elements of b into result

Mergesort recursive: merge two lists

- If one of the two lists **`a`** and **`b`** is empty, output the other
- Otherwise, there is a first (smallest) element in each of the two lists
- From these two elements, take the smallest one
- Behind it, put the list by merging the remaining elements

Code: merge two lists

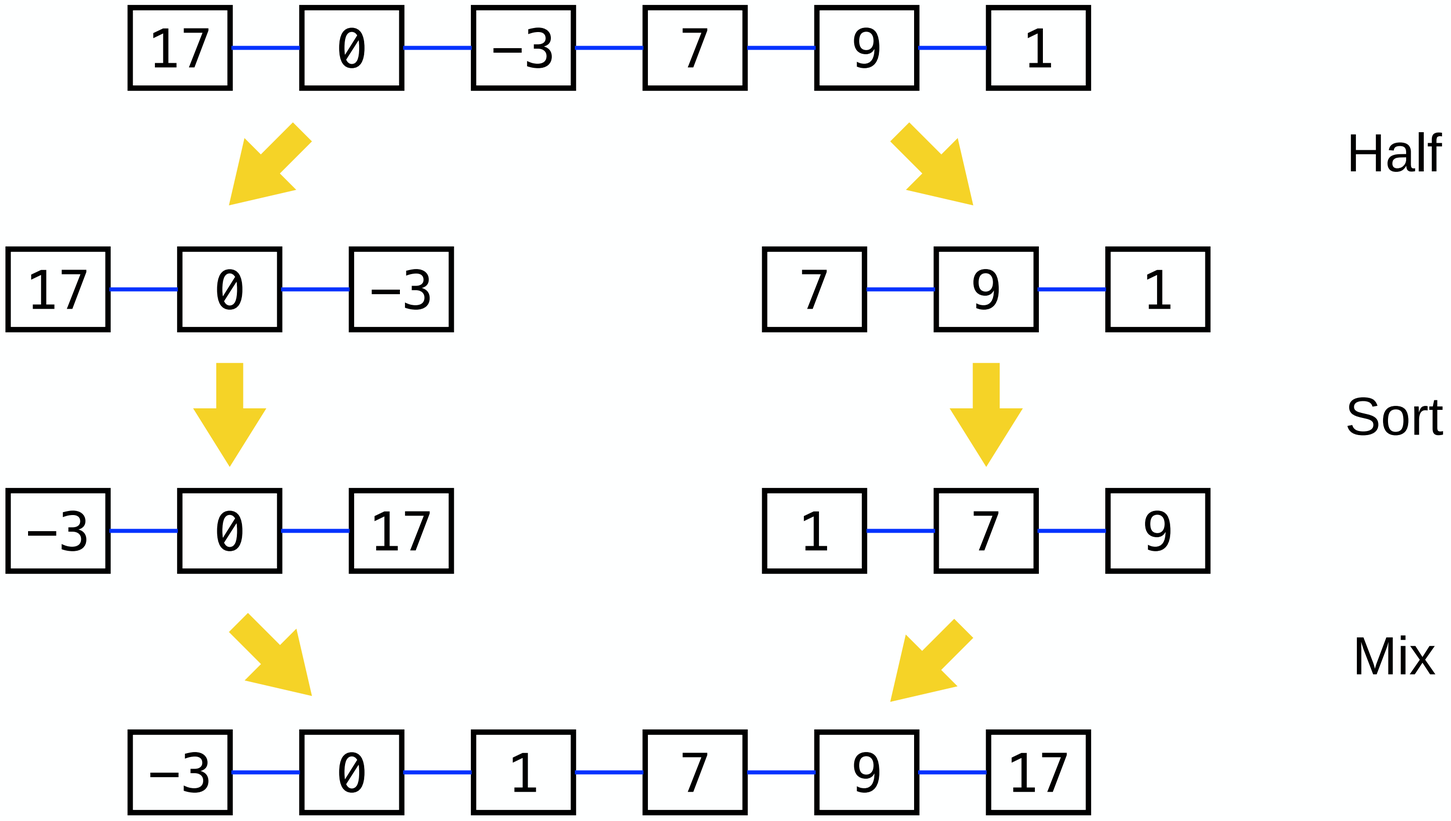
```
public static <T extends Comparable<T>> List<T> mergeRecursive(List<T> a, List<T> b) {
    if (a == null || a.isEmpty()) {
        return b;
    }
    if (b == null|| b.isEmpty()) {
        return a;
    }
    List<T> newList = new ArrayList<>();
    if (a.get(0).compareTo(b.get(0)) < 0) {
        newList.add(a.remove(0));
    } else {
        newList.add(b.remove(0));
    }
    newList.addAll(mergeRecursive(a, b));
    return newList;
}
```

Terminating condition

Direct recursive method calls

Sort by mixing

- Divide list to be sorted into two partial lists
- Sort each half separately
- Mix the results
- Example



Code: half

```
public static <T extends Comparable<T>> List<T> leftHalf(List<T> list) {
    List<T> leftHalf = new ArrayList<>();
    for (int i = 0; i < list.size() / 2; i++) {
        leftHalf.add(list.get(i));
    }
    return leftHalf;
}

public static <T extends Comparable<T>> List<T> rightHalf(List<T> list) {
    List<T> rightHalf = new ArrayList<>();
    for (int i = list.size() / 2; i < list.size(); i++) {
        rightHalf.add(list.get(i));
    }
    return rightHalf;
}
```

You can find the whole code including usage examples on https://gist.github.com/krusche/d18e381275016e2e74e98157831d78ac

Code: sort

```
public static <T extends Comparable<T>> List<T> sort(List<T> list) {
    if (list == null || list.size() <= 1) {
        return list;
    }
    List<T> left = leftHalf(list);
    List<T> right = rightHalf(list);
    left = sort(left);
    right = sort(right);
    return mergeRecursive(left, right);
}
```

Terminating condition

Direct recursive method calls

You can find the whole code including usage examples on https://gist.github.com/krusche/d18e381275016e2e74e98157831d78ac

W12E04 - Fibonacci Sequence

Start exercise

Easy

Not started yet.

Due by tonight

10 min

3 pts

- **Problem statement**
  - Fibonacci sequence: `0 1 1 2 3 5 8 13 21 34 55 ...`
  - The **`n-th`** element is the sum of the two previous elements
  - Calculate the **`n-th`** element of the fibonacci sequence using a recursive function
  - Examples
    - `fibonacci(2)  = 1`
    - `fibonacci(3)  = 2`
    - `fibonacci(7)  = 13`
    - `fibonacci(10) = 55`
  - Optimize the calculation with an iterative version

Fibonacci performance characteristics

- Exponential complexity: $O(2^n)$
- Bad implementation: lots of repeated and unnecessary calculations!
- **`fibonacci(50)`** cannot be computed in a reasonable amount of time using the recursive function

```
                          fib(5)
                     /                 \
               fib(4)                 fib(3)
             /        \               /       \
         fib(3)      fib(2)         fib(2)   fib(1)
        /    \       /    \        /      \
  fib(2)   fib(1)  fib(1) fib(0) fib(1) fib(0)
  /     \
fib(1) fib(0)
```

Iterative solution

- Complexity: **`O(n)`**

```java
public static long fibonacciIterative(int n) {

    long[] fibNumbers = new long[n + 2];

    fibNumbers[0] = 0;
    fibNumbers[1] = 1;

    for (int i = 2; i <= n; i++) {

        fibNumbers[i] = fibNumbers[i - 1] + fibNumbers[i - 2];
    }

    return fibNumbers[n];
}
```

Declare an array to store fibonacci numbers (2 additional elements to handle the cases **`n = 0`** and **`n = 1`**)

0th and **1st** number of the series are **0** and **1**

Add the previous **2 numbers** in the series and store it

Next steps

- **Tutor group** exercise
  - T12E01 - Into the Seal Matrix (1999)
- **Homework** exercise
  - H12E01 - TUM Seal Triangle
- **Project work**
  - Implement the game
- Read the following articles
  - https://www.baeldung.com/java-recursion
  - https://introcs.cs.princeton.edu/java/23recursion

→ Due by **Wednesday, January 21, 13:00**

Summary

- Recursion is the technique of **invoking a function within itself**
- It provides a way to **break complicated problems down into simple ones** which are easier to solve
- Recursion may be **difficult to understand**
- The best way to figure out how it works is to **experiment with it in multiple examples**
- There are two main requirements of a recursive function
  - **A terminating condition:** the function returns a (simple) value when a certain condition is satisfied, without a further recursive call
  - **The recursive call:** the function calls itself with an input which is a step closer to the terminating condition
- Sometimes, iteration is the significantly faster approach (e.g. fibonacci)

References

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- https://www.geeksforgeeks.org/merge-sort
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- https://books.trinket.io/thinkjava2/chapter8.html
- https://codingbat.com/java/Recursion-1
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- https://mathworld.wolfram.com/KochSnowflake.html
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- https://www.baeldung.com/java-binary-search